RZera User Manual

CSNS Data Analysis Group

RZera User Manual

The purpose of this manual is to explain to users the workflow for data reduction using RZera, and to guide users step by step through the data reduction process. If the data in your paper have been processed using RZera, please cite reference [1]. If you use and adopt the prediction results of the AI module "AIndex" in RZera in your paper, please cite reference [2]. If you use the small-angle scattering module, please cite reference [3]. If you encounter any issues during the use of the diffraction reduction function, please contact wanghao96@ihep.ac.cn. For issues during the use of the small-angle reduction function, please contact zhongjiajun@ihep.ac.cn. For collaboration inquiries, please contact durong@ihep.ac.cn.

Reference：

[1] Wang H, Liu Z. Y, Guo H. C, Peng X. G, Xu J. P, Yin W, Zhang J. R, Du R. RZera: A customized and user-friendly software for the total scattering data reduction at CSNS[J]. ***Nucl. Instrum. Methods Phys. Res., Sect. A***, 2024,1066: 169607.

[2] Wang H, Zhong J, Li Y, Zhang J, Du R. Designing a Dataset for Convolutional Neural Networks to Predict Space Groups Consistent with Extinction Laws[J]. ***Applied Crystallography***, 2025, 58(3).

[3] Jiang H*, Zhong J*, Chen C, Zhang J, Du R. sansRZ: A python-based data reduction tool for the time-of-flight SANS instrument at CSNS[J]. ***Nucl. Instrum. Methods Phys. Res., Sect. A***, 2026, 1085: 171283.

Contents

1. Instructions for Using Powder Diffraction Spectrometer Software

1.1 Data Processing Workflow for Powder Diffraction in RZera

RZera splits the data reduction workflow for neutron powder diffraction into several sequential steps, as shown in Figure 1.1-1. In the first step, the data of the sample, sample cell, V/VNi, V/VNi cell, and empty background are each subjected to time focusing, which converts the two-dimensional histogram data into one-dimensional I-d curves. RZera provides a dedicated "Time Focusing" page for this operation. In the second step, the one-dimensional curves are used for background subtraction and vanadium normalization to obtain the reduced I-d curve of the sample. This operation is handled by the "Reduction" page in RZera. Note that some spectrometers do not require empty background data; in such cases, the data processing steps related to the empty background can be omitted. In the subsequent figures, the background refers specifically to the cell data. After that, users can continue to process the data to obtain the PDF. In RZera, there are two methods for obtaining the PDF: one is the Merge + Fourier transform method, and the other is the Hermite polynomial fitting method.

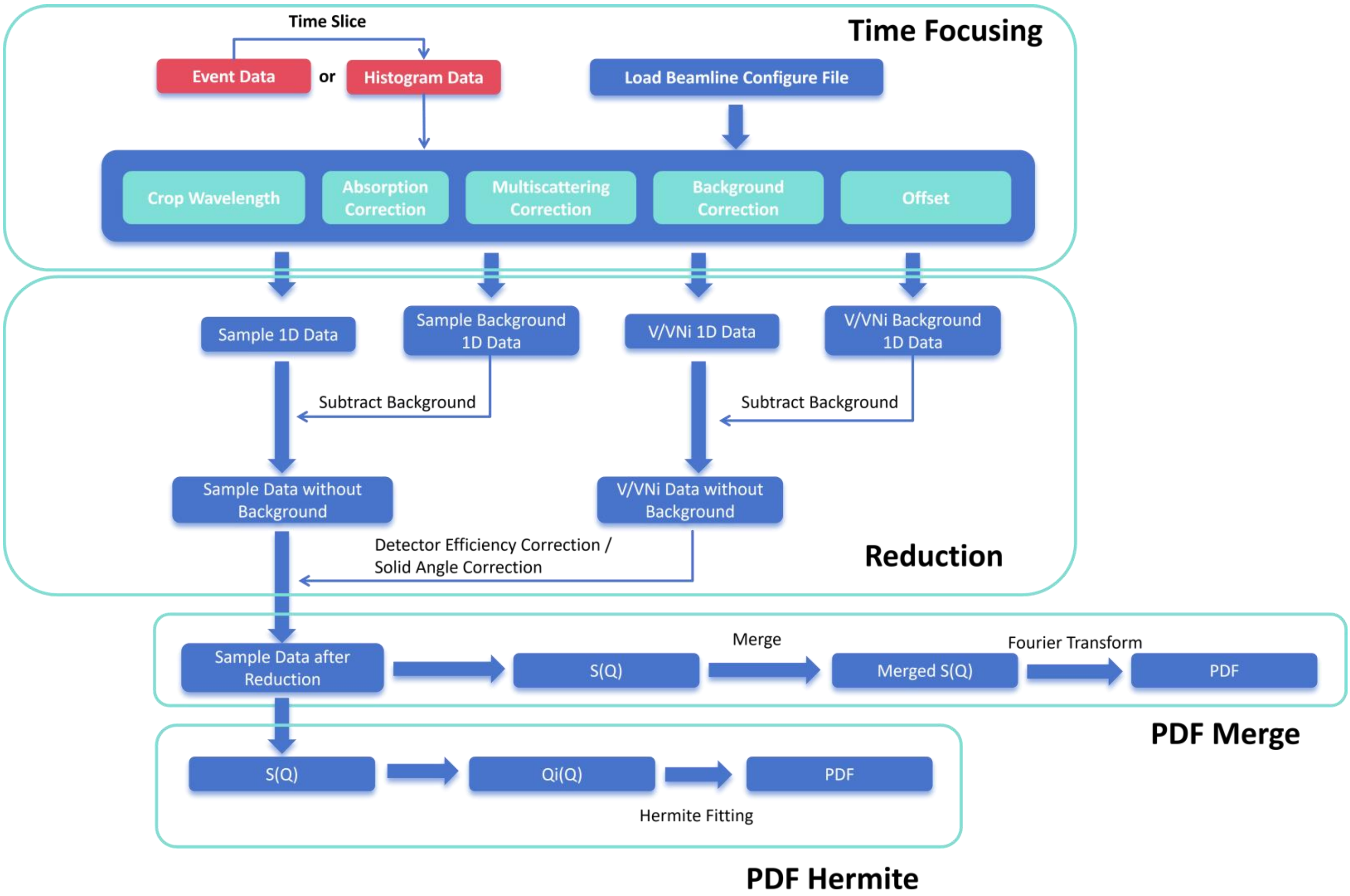


Figure 1.1-1 Data reduction workflow in RZera without empty background subtraction. In the figure, "background" refers to the cell. If empty background subtraction is required, the sample, V/VNi, and cell data must first be subtracted once by the empty background data.

In the above reduction workflow, almost every step supports visualization, allowing users to conveniently check the changes in the data throughout the entire process. However, because the steps are divided in considerable detail, the operation can be rather tedious. To address this, RZera also provides a quick reduction function: once the relevant parameters are properly configured, batch data reduction can be achieved with a single click, as shown in Figure 1.1-2. First, users still need to perform time focusing on the V/VNi data and its background data, but these two datasets can be reused across a beamtime experiment. Then, RZera provides a "Quick Reduction" page, which accepts the time-focused V/VNi and V/VNi background, as well as the pre-time-focusing sample and sample background (post-time-focusing background data are also acceptable) as inputs. After the user has set all parameters on the interface, the reduced sample data can be obtained with one click. RZera also offers a configuration saving feature: when data from different samples are collected next time, users only need to import the saved configuration, replace the sample data, and then perform the one-click reduction again.

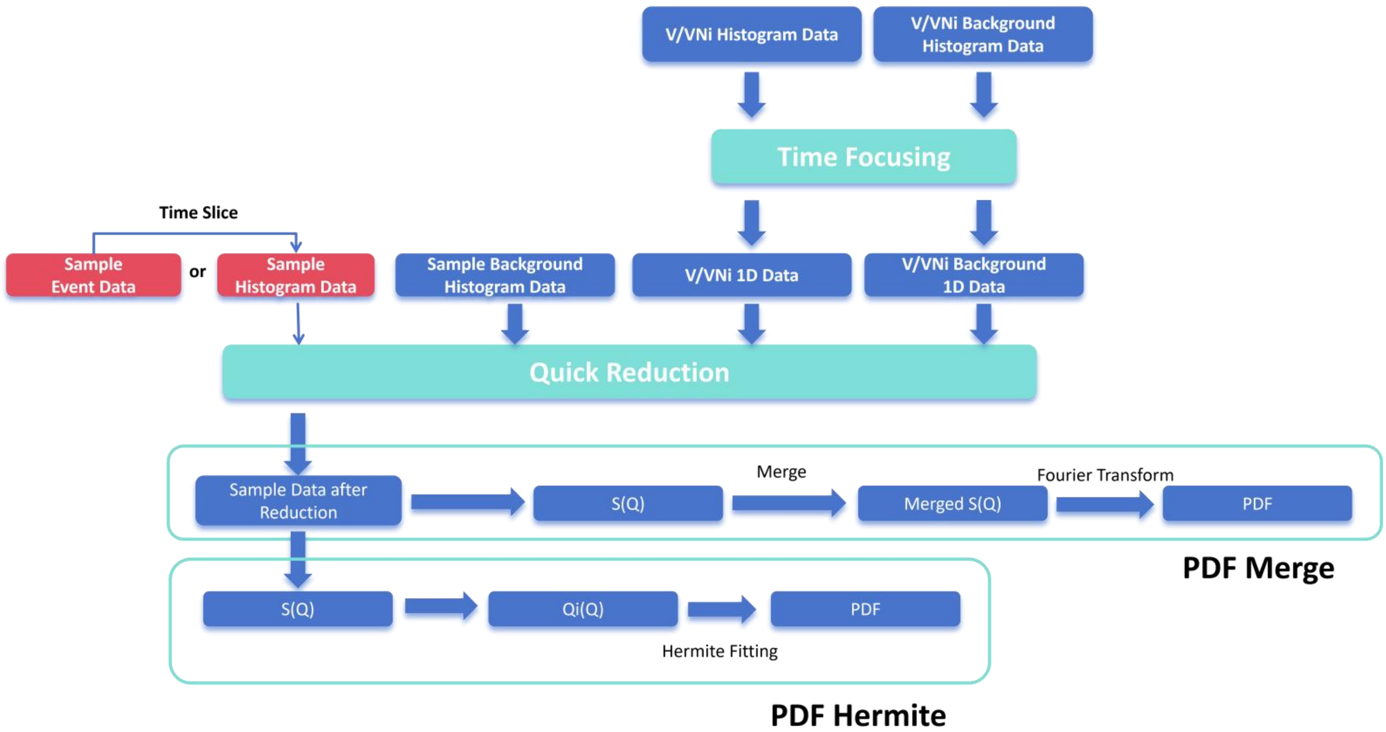


Figure 1.1-2 Quick data reduction workflow in RZera.

1.2 Data Reduction Using Histogram Data

1.2.1 File Preparation

Users need to prepare four files, namely the nxs files (provided by the spectrometer) for the V or VNi data, the V or VNi background, the sample data, and the sample background. The V or VNi background and the sampl e background may be the same dataset; in that case, only three files need to be prepared. As shown in Figures 1.2-1 and 1.2-2, there are three folders named after RUN numbers, which contain the nxs files for the background, V, and Si standard sample, respectively.

Figure 1.2-1 Folders storing nxs files, with folder names corresponding to RUN numbers. The RUN number is used to label each data collection experiment.

Figure 2 Data inside a folder named by the RUN number. The nxs file is "detector.nxs" as shown in the figure, which stores the histogram data and some metadata of the experiment. There may be other data files in this folder as well, but this chapter, "Data Reduction Using Histogram Data," only uses the "detector.nxs" file.

1.2.2 Entering the RZera data reduction interface for the corresponding spectrometer

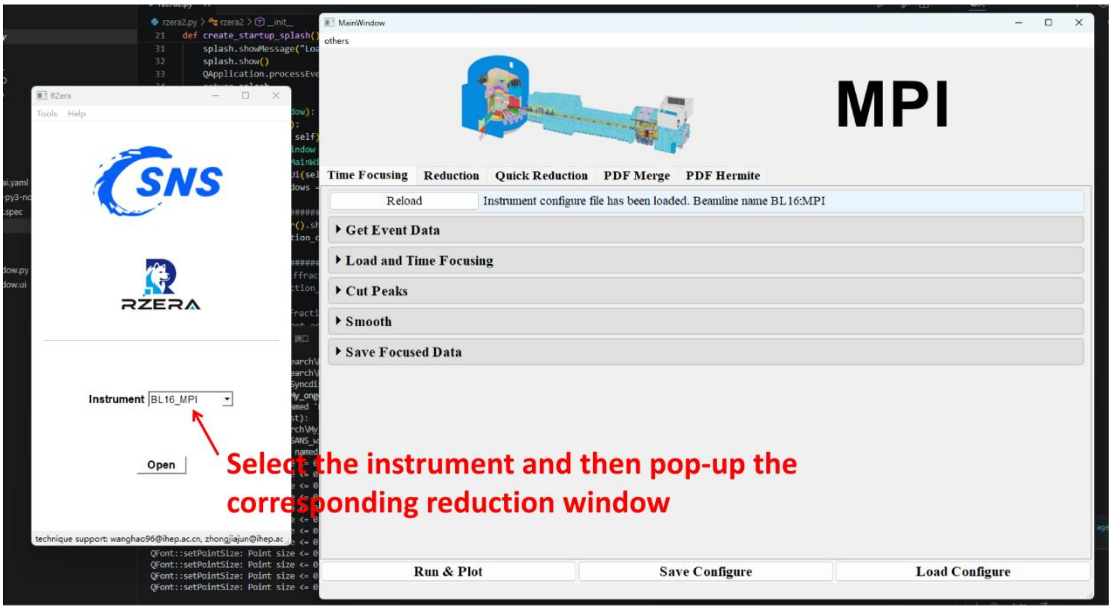


Figure 1.2-3 How to enter the data reduction window for a specified spectrometer.

As shown in Figure 1.2-3, after opening RZera, there is a drop-down box for selecting a spectrometer. Once the spectrometer used in the experiment is selected, the corresponding data reduction window for that spectrometer will pop up.

1.2.3 Performing Time Focusing on the V/VNi Data

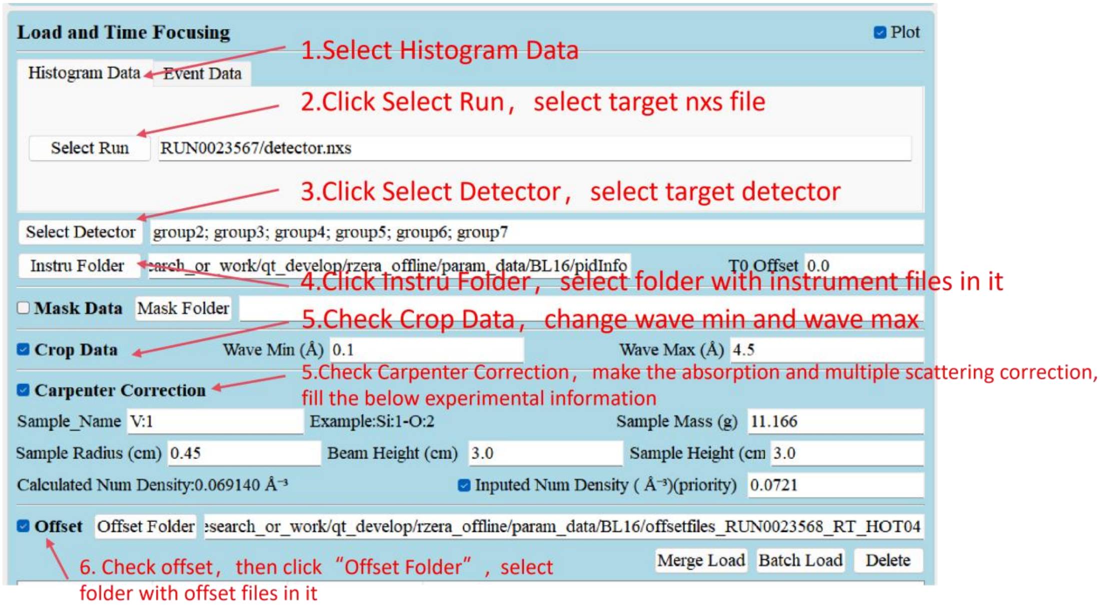


Figure 1.2-4 Information input steps in "Load and Time Focusing"

In the pop-up reduction window, there is a Time Focusing page, where the first module to be operated is "Load and Time Focusing", as shown in Figure 4. When importing data following the instructions in Figure 1.2-4, several points should be noted:

(1) Regarding the "Batch Load" option: Users can import data from multiple Runs simultaneously. If "Batch Load" is checked, RZera will perform time focusing on each of the imported datasets separately. However, if it is not checked, RZera will merge the data from these Runs and then perform time focusing on the merged data.

(2) Regarding "Instru Folder", RZera will provide this.

(3) Regarding "Sample_Name" under "Carpenter Correction": Enter the ratio according to the actual molar fraction of the V:Ni sample. If the exact number density value is known, instead of using the calculated value, you may check "Inputed Num Density" and manually enter the accurate value.

(4) Regarding "Offset Folder", RZera or the spectrometer team can provide this.

After completing the information input, click "Load", and the software will begin loading the data and performing time focusing based on the entered information. The loaded data will then be saved in the list at the bottom of this module, as shown in Figure 1.2-5.

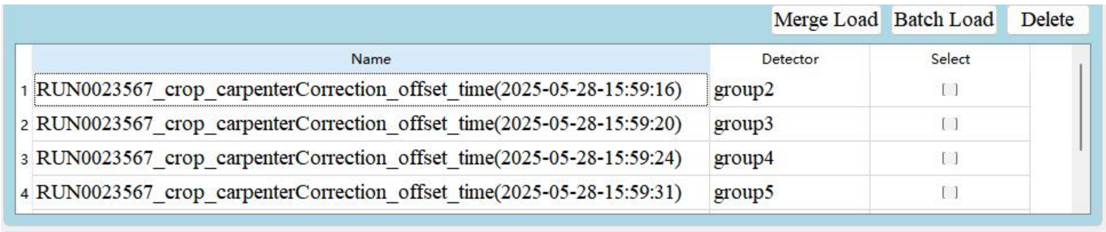


Figure 1.2-5 Data obtained after loading.

After the data are loaded, this page also provides peak removal and smoothing functional modules, located below the "Load and Time Focusing" module, as shown in Figure 1.2-6. These modules are collapsed by default and are used to remove unwanted diffraction peaks from the data. For vanadium, we need to remove all of its diffraction peaks.

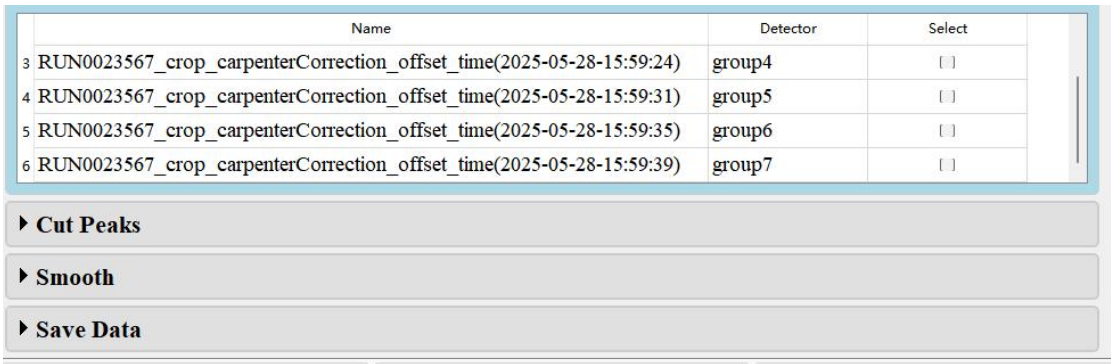


Figure 1.2-6 Peak removal and smoothing functional modules.

After clicking "Cut Peaks", the module will expand, as shown in Figure 1.2-7. This module consists of two parts: the upper part is used to automatically calculate the positions of diffraction peaks to be removed based on a CIF file, and the lower part is for manually locating diffraction peak positions.

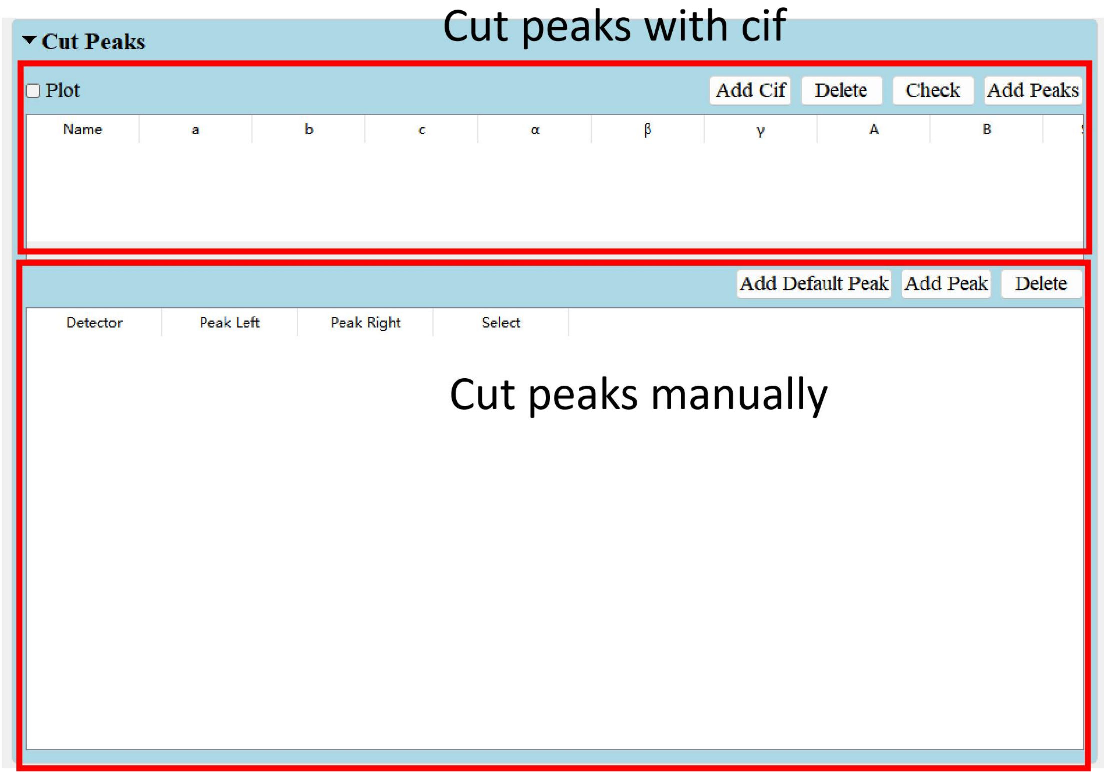


Figure 1.2-7 The "Cut Peaks" functional module.

RZera has built-in default peak positions for vanadium. Simply click "Add Default Peak" in the manual peak removal section to automatically import the V peak positions, as shown in Figure 1.2-8.

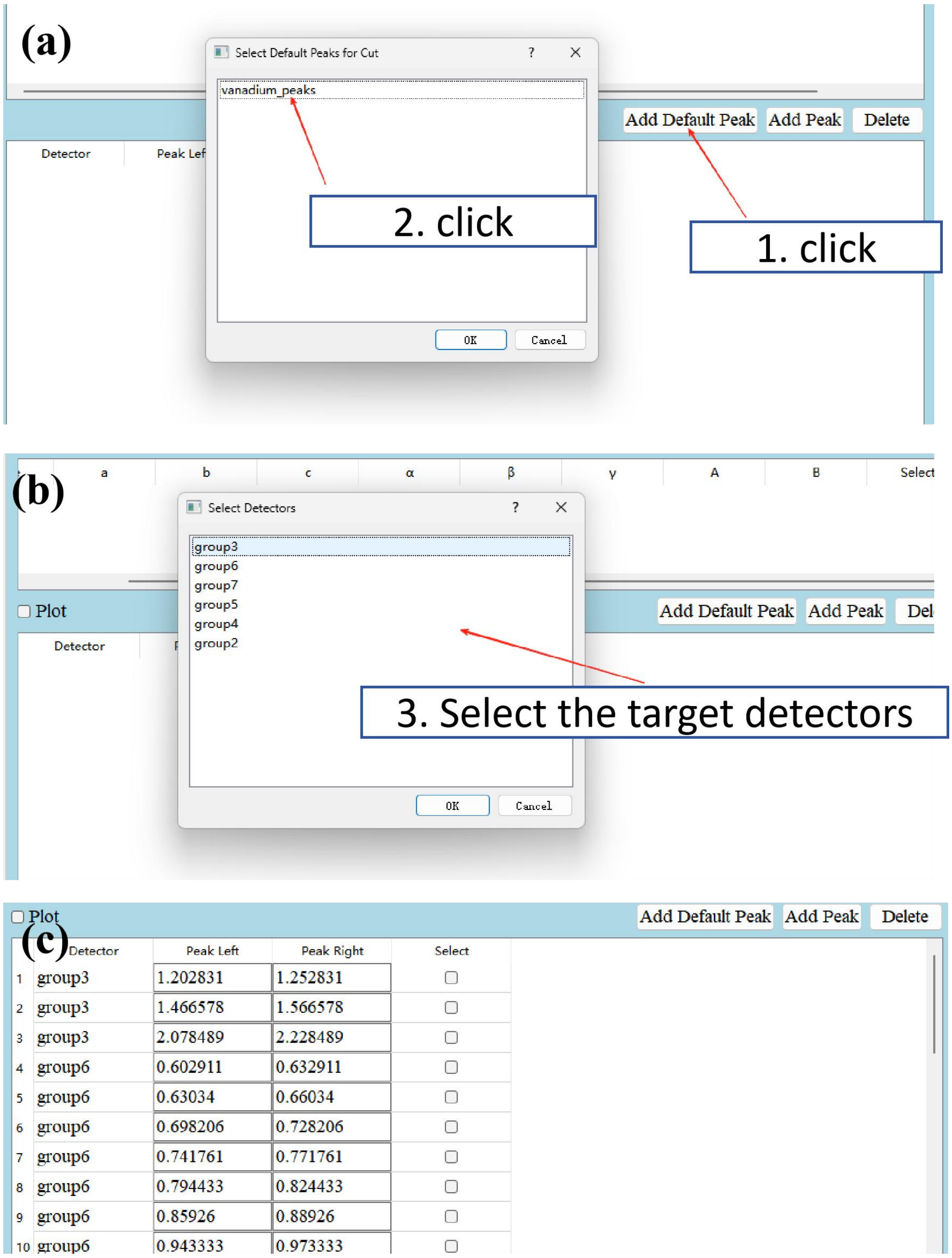


Figure 1.2-8 Importing default V peaks. (a) After clicking the button, select the type of default peaks to import. (b) Select the detector for which the peak information is to be imported. The detectors in this list are consistent with those selected during the previous data loading step. (c) The peak position information displayed after import.

After importing, check the "plot" boxes in both the "Load and Time Focusing" module and the "Cut Peaks" module, then click the "Run & Plot" button at the bottom of the entire page. The "Run & Plot" button will execute the enabled functional modules in order from top to bottom. Additionally, if the "plot" box in a module is checked, the data processed by that module will be added to the plotting window and the plotting window will pop up, enabling data visualization, as shown in Figure 1.2-9.

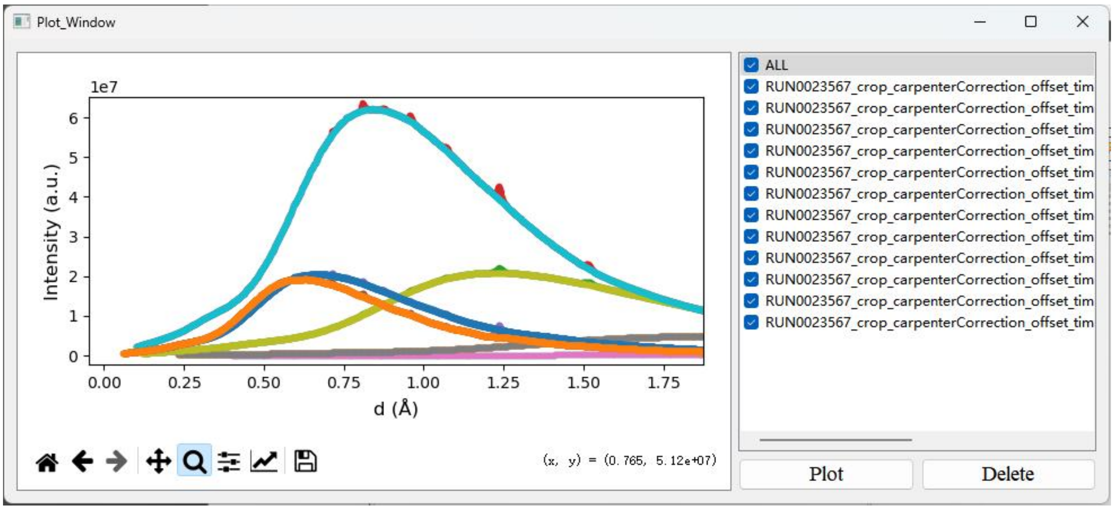


Figure 1.2-9 Comparison of data before and after peak removal. The reason for this comparison is that the software executes modules from top to bottom: first, "Load and Time Focusing" is executed, and since the "plot" box in this module is checked, the data before peak removal are saved in the plotting window. Then, "Cut Peaks" is executed, and after peak removal, since the "plot" box in this module is also checked, the data after peak removal are saved in the plotting window.

After peak removal, the removed portions are filled with straight lines. RZera provides the "Smooth" module (Figure 1.2-6) to smooth the data, and its expanded view is shown in Figure 1.2-10. Clicking "Add Default Smooth" within this module allows you to apply the built-in smoothing parameters of RZera with one click.

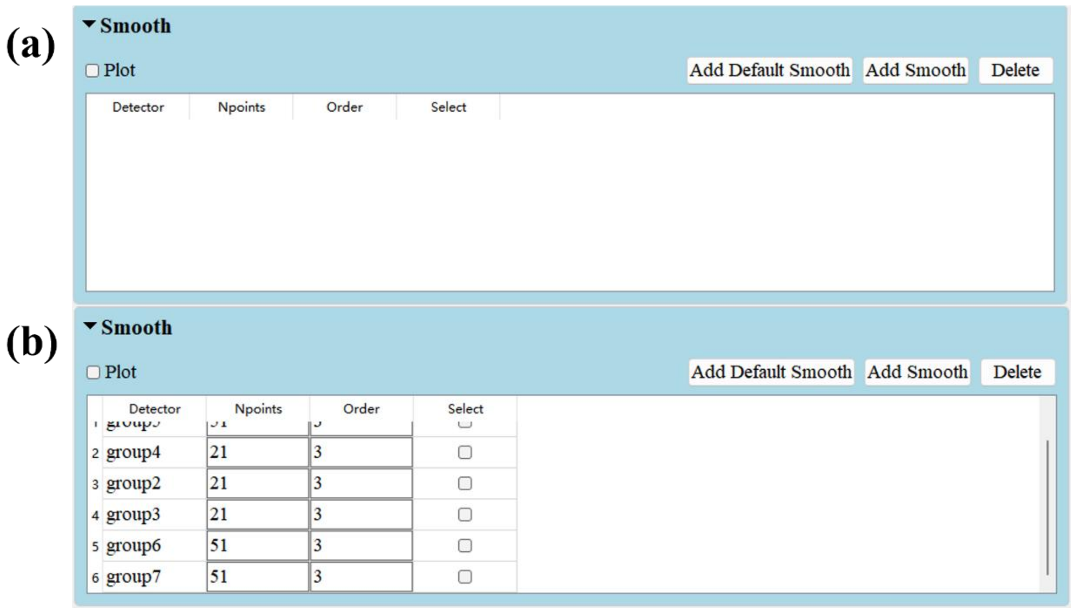


Figure 1.2-10 (a) The expanded interface of the "Smooth" module. (b) The interface after clicking "Add Default Smooth".

After checking the "plot" box in the "Smooth" module as well, click the "Run & Plot" button to achieve a comparison of the data before and after smoothing, as shown in Figure 1.2-11.

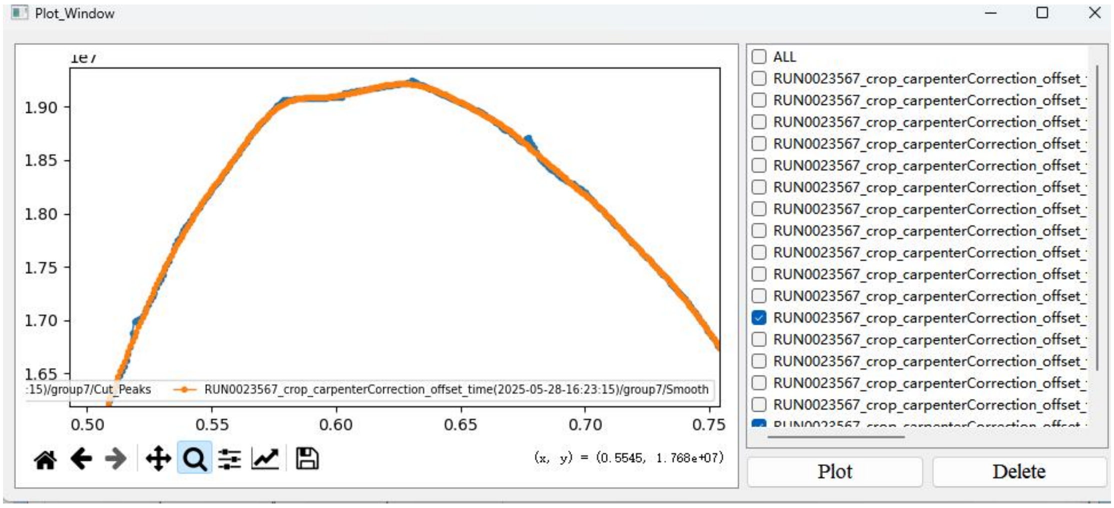


Figure 1.2-11 Comparison of data from detector group 7 before and after smoothing. It can be seen that the data are significantly smoother after smoothing.

After smoothing, expand the final "Save Data" module (as shown in Figure 1.2-12) to prepare for saving the processed data. After configuring the "Save Data" module, click "Run & Plot" again, and the processed data will be saved to the specified path.

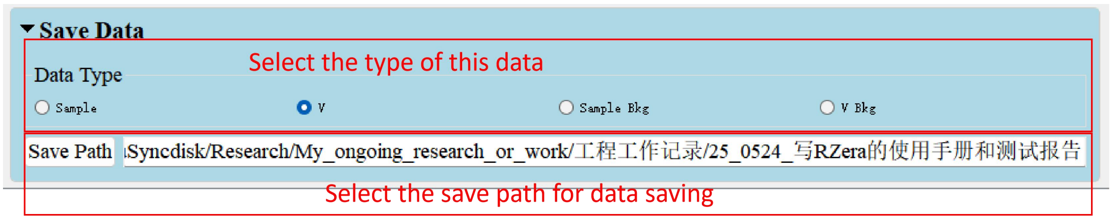


Figure 1.2-12 The interface of the "Save Data" module. The "Data Type" must be consistent with the type of data currently being processed. In this example, we are processing V or VNi sample data, so "V" should be selected. Then click "Save Path" to specify the save location.

The processed file is a ".nc" file, which stores the processed data and the processing history internally, as shown in Figure 1.2-13.

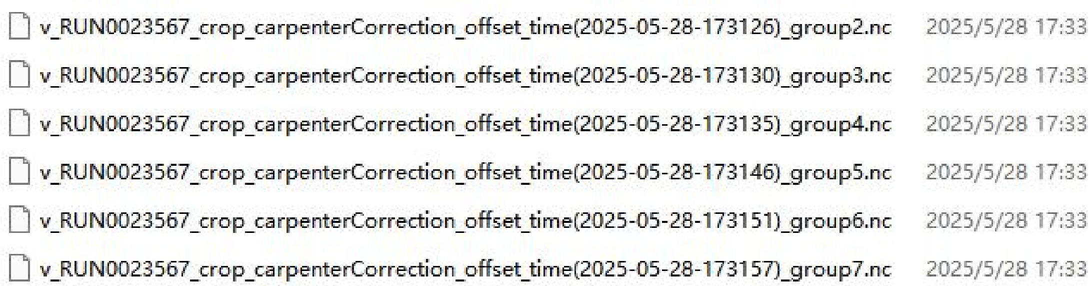


Figure 1.2-13 Saved Data

1.2.4 Performing Time Focusing on the V/VNi Background

The operation procedure for this step is the same as in the previous step (Step 3), but there are several differences or points to note, as follows:

(1) The nxs file to be loaded should be replaced with the background

corresponding to V/VNi.

(2) If Carpenter Correction was applied to the V/VNi data during "Load and Time Focusing", it is recommended to apply exactly the same Carpenter Correction to the background data as well, in order to cancel out its effect during the subsequent background subtraction.

(3) Since the background may not contain diffraction peaks, the Cut Peaks and Smooth steps may not be necessary, depending on the actual situation.

(4) When saving the data at the end, "Data Type" should be set to "V Bkg".

The final saved data are shown in Figure 1.2-14.

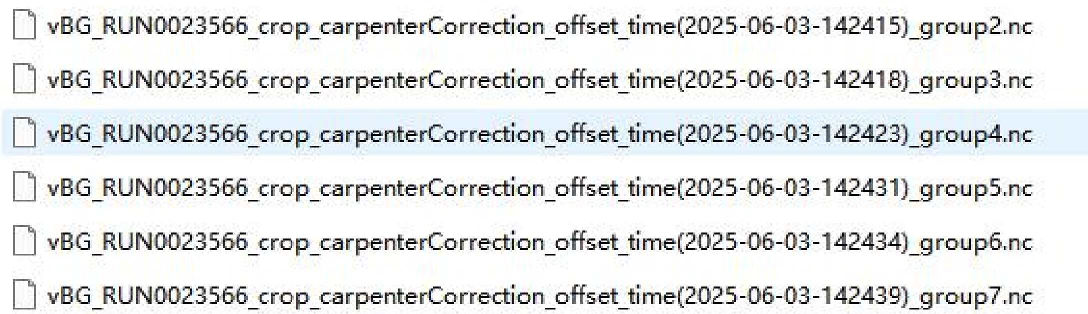


Figure 1.2-14 The saved V/VNi background data.

1.2.5 Performing Time Focusing on the Sample

The operation procedure for this step is the same as Step 3, but there are several differences or points to note, as follows:

(1) The nxs file to be loaded should be replaced with that of the sample.

(2) Carpenter Correction is not necessarily required for the sample. If only diffraction data are needed, it can be skipped. However, if further processing to obtain the PDF is required, then it is needed. If the correction is to be applied, it is necessary to ensure that the sample is either a cylinder or a powder contained in a cylindrical container, and that all relevant input parameters are entered correctly; otherwise, the correction may be inaccurate.

(3) Since the sample may not contain extraneous diffraction peaks, the Cut Peaks and Smooth steps may not be needed, depending on the specific situation.

(4) When saving the data at the end, "Data Type" should be set to "Sample".

The final saved data are shown in Figure 1.2-15.

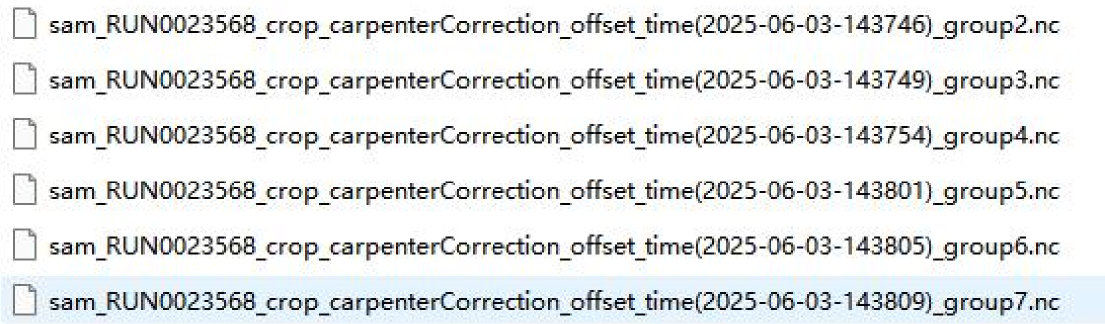


Figure 1.2-15 The saved sample data.

1.2.6 Performing Time Focusing on the Sample Background

The operation procedure for this step is the same as Step 3, but there are several differences or points to note, as follows:

(1) The nxs file to be loaded should be replaced with that of the sample background.

(2) If Carpenter Correction was applied to the sample during "Load and Time Focusing", it is recommended to apply exactly the same Carpenter Correction to the background data as well, in order to cancel out its effect during the subsequent background subtraction.

(3) Since the background may not contain diffraction peaks, the Cut Peaks and Smooth steps may not be necessary, depending on the actual situation.

(4) When saving the data at the end, "Data Type" should be set to "Sample Bkg".

The final saved data are shown in Figure 1.2-16.

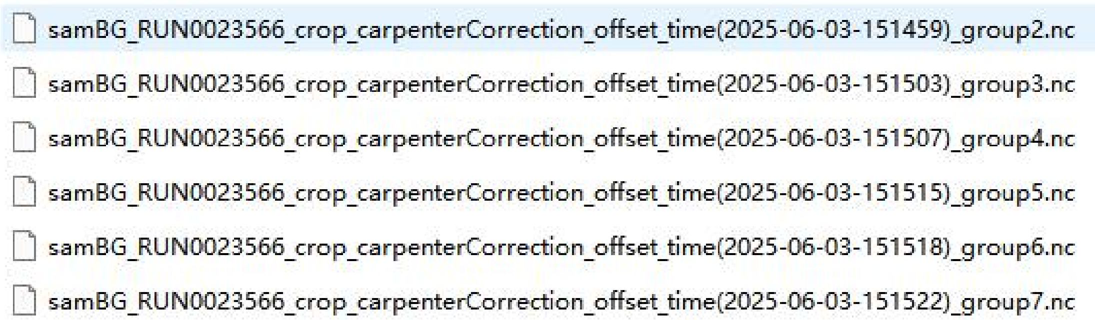


Figure 1.2-16 The saved sample background data.

1.2.7 Using the data obtained in Sections 1.2.3–1.2.6 to obtain the reduced diffraction pattern.

After completing Steps 3–6, we have obtained the time-focused data for four types: sample, sample background, V/VNi, and V/VNi background. Next, we need to switch the tab page from "Time Focusing" to "Reduction". The interface of the Reduction page is shown in Figure 1.2-17.

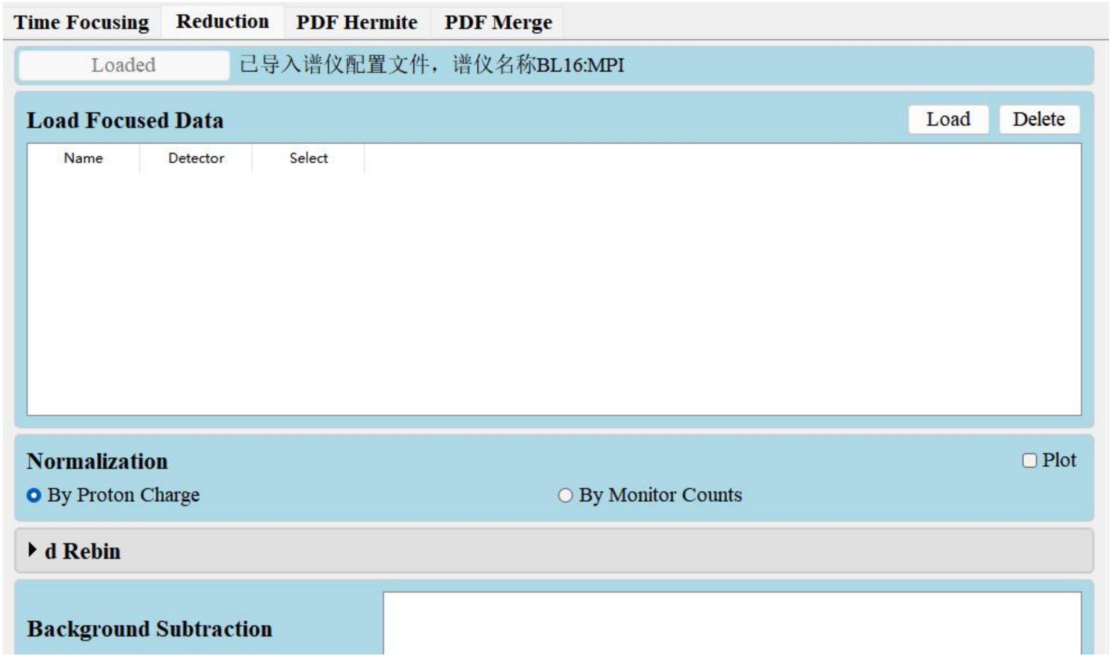


Figure 1.2-17 The "Reduction" tab page.

First, we operate the "Load Focused Data" module. Click "Load" to import all the data obtained in Steps 3–6, as shown in Figure 1.2-18.

Figure 1.2-18 (a) Schematic diagram of the "Load Focused Data" module operation. (b) The "Load Focused Data" module after data loading.

Next, consider the "Normalization" module, which offers two options: "By Proton Charge" and "By Monitor Counts". These refer to normalizing the data by proton charge and by monitor counts, respectively. In theory, either choice is correct; however, there may be cases where a monitor was not used during the experiment, in which case selecting "By Monitor Counts" will cause the program to report an error. Therefore, the appropriate option should be chosen based on the actual experimental conditions.

Next, expand the "d_Rebin" module and click "Add Default Rebin". After selecting the detectors to be processed in the pop-up window, the module will load the default d_rebin parameters for each detector, as shown in Figure 1.2-19. After loading the default parameters, users can also modify the d_rebin parameters directly in the interface.

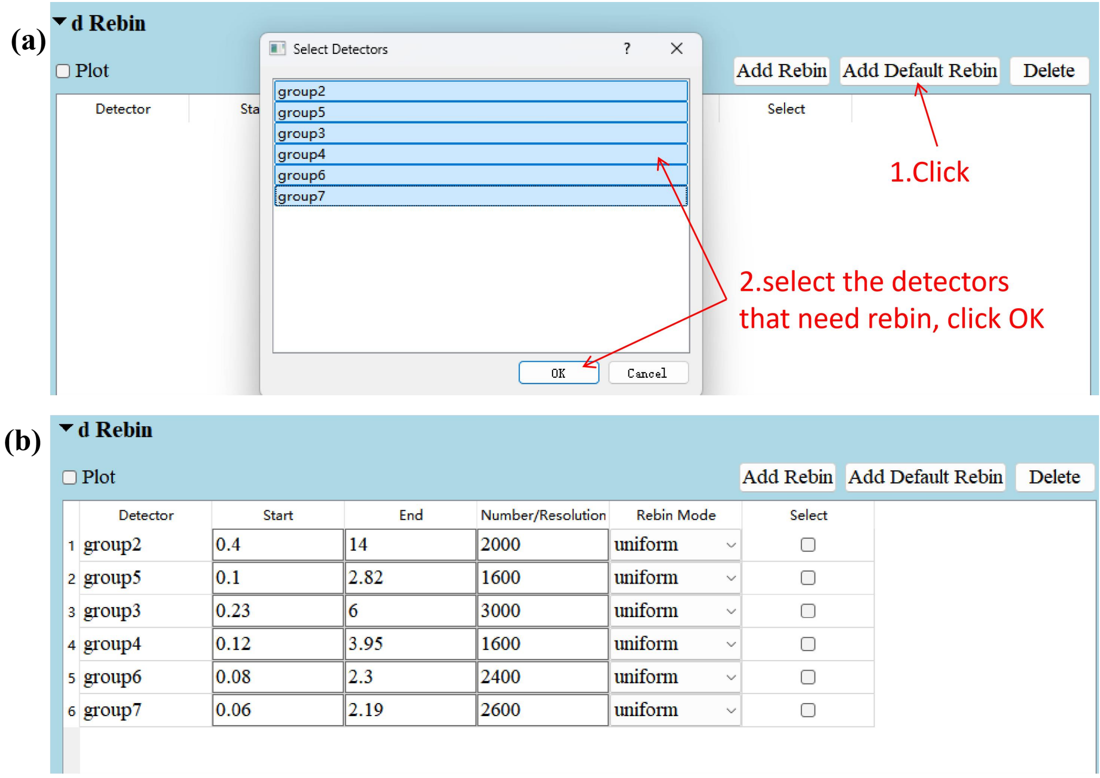


Figure 1.2-19 (a) The workflow for operating the "d_Rebin" module. (b) The "d_Rebin" module after loading the default parameters.

Next, operate the subsequent "Background Subtraction" module as shown in Figure 1.2-20. This module is dedicated to background subtraction. It can be noted that there are multiple consecutive "Background Subtraction" modules in the interface, which are used for subtracting the background from the sample and V/VNi data respectively (there is no required order for the background subtraction). Click "Load Sample" to load the sample or V/VNi data, and then click "Load Background" to load the corresponding background. After loading, the module provides real-time display of the data after background subtraction. The "Background Scale" in the module refers to the scale factor multiplied to the background before subtraction. Sliding the "Background Scale" slider allows adjustment of this factor, and the changes in the data after applying the factor are reflected in real time in the plot on the right. The default scale factor is 1, but there may be cases where the background is higher than the data; in such cases, the scale factor must be adjusted to ensure that all data remain positive after background subtraction. For guidance on which value to choose as the scale factor, please consult the spectrometer scientist for the corresponding instrument.

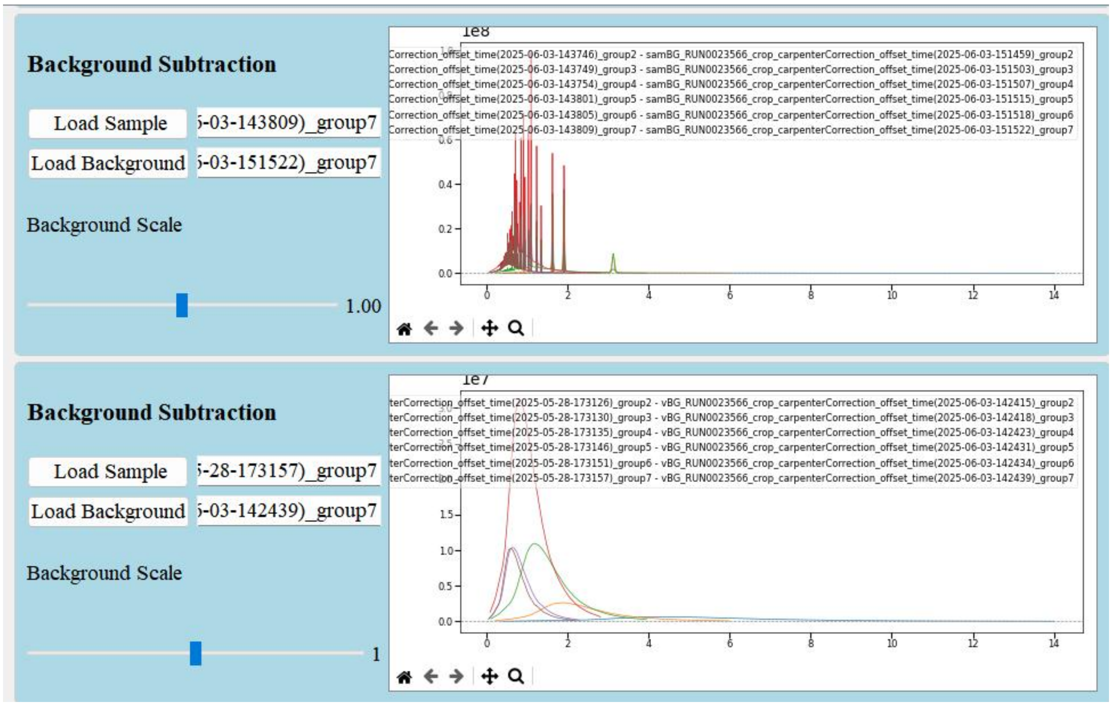


Figure 1.2-20 The interface after loading the sample, V/VNi, and their corresponding backgrounds. After the data are loaded, the right panel will display the background-subtracted data. If the data do not appear, you can click the "home" button (the small black house icon) in the lower-left corner of the plotting area, and the data will be displayed normally.

Next, operate the subsequent "Division" module. Click "Load Sample" to import the sample data, and click "Load Calibration" to import the V/VNi data. The data imported here will be divided after having undergone background subtraction from the preceding modules. The module after data import is shown in Figure 1.2-21.

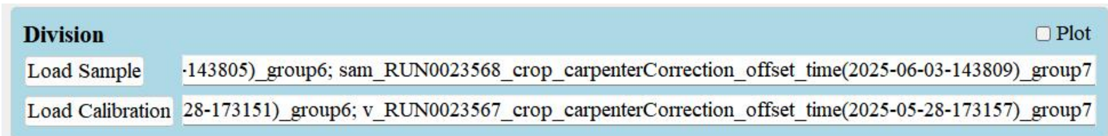


Figure 1.2-21 The "Division" module after importing the sample and V/VNi data.

Next is a skippable module, "AIndex", which is developed based on research results published in 2025 by the China Spallation Neutron Source data analysis group. Its function is AI-based prediction of space groups and unit cell parameters. Users only need to import the reduced data to predict the space group and unit cell parameters with one click, and can easily check whether the prediction results are reliable.

First, click the "Load Pattern" button to import the reduced sample data, as shown in Figure 1.2-22 (a)(b). Then, click the "Prediction" button, and the software will call the trained deep neural network to provide prediction results for the crystal system, space group, and unit cell parameters corresponding to each diffraction pattern, which are displayed in the list panel on the right, as shown in Figure 1.2-23. The panel contains two columns: the first column shows the prediction results, and the second column shows the confidence level of each prediction. By selecting a prediction result in the list panel and clicking the "Check" button, AIndex will calculate the diffraction peak positions using the predicted results and display them together with the diffraction pattern, as shown in Figure 1.2-23, allowing users to conveniently verify the accuracy of the prediction.

There are certain limitations to using this module, as well as specific requirements for the data: (1) Currently, AIndex can only perform predictions for cubic, tetragonal, trigonal, and hexagonal crystal systems. (2) The data must be from a single-phase sample; otherwise, correct prediction cannot be performed. (3) The unit cell parameters of the sample should be within the range of 3–15 Å; otherwise, correct prediction cannot be performed. (4) The data should ideally cover the range 0.5–10 Å, or it must be ensured that no diffraction peaks exist in the uncovered regions; otherwise, the prediction results are likely to be inaccurate.

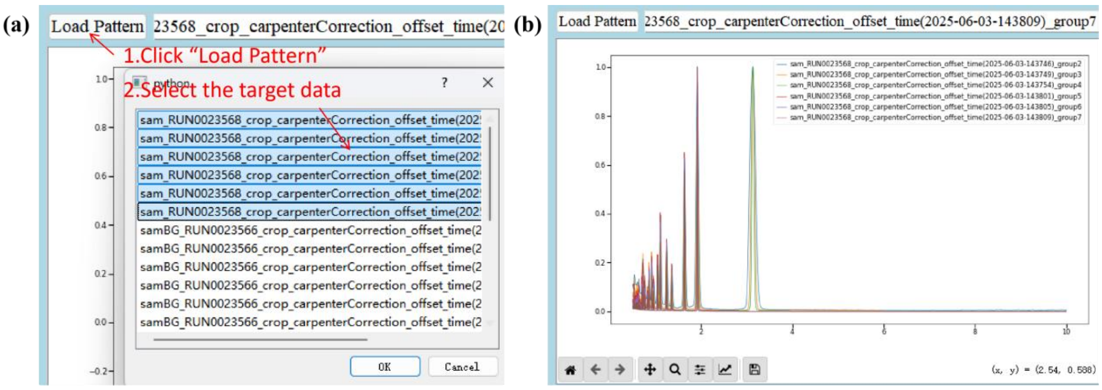


Figure 1.2-22 (a) The workflow for importing data into "AIndex". (b) After data import, AIndex reorganizes the data into a diffraction pattern consisting of 4196 equally spaced points over the range 0.5–10 Å and displays it. This reorganization is intended to match the input format required by the neural network and is used only internally within this module; it does not affect the final reduction results.

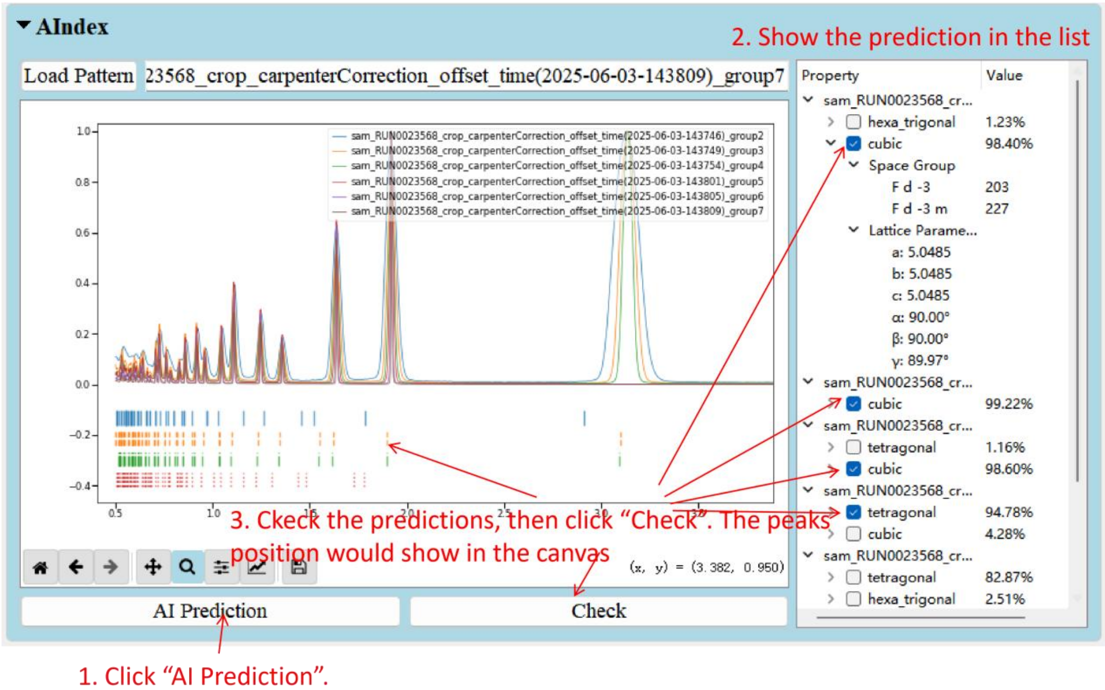


Figure 1.2-23 The workflow for predicting space group and unit cell parameters using AIndex with the assistance of AI.

Then, operate the "Save Reduced Data" module. This module is very simple: after expanding it, just click "Save Path" to specify the save location for the data. Finally, click "Run & Plot", and the software will process the modules one by one from top to bottom. The processed data will then be saved to the path specified in the "Save Reduced Data" module, as shown in Figure 1.2-24.

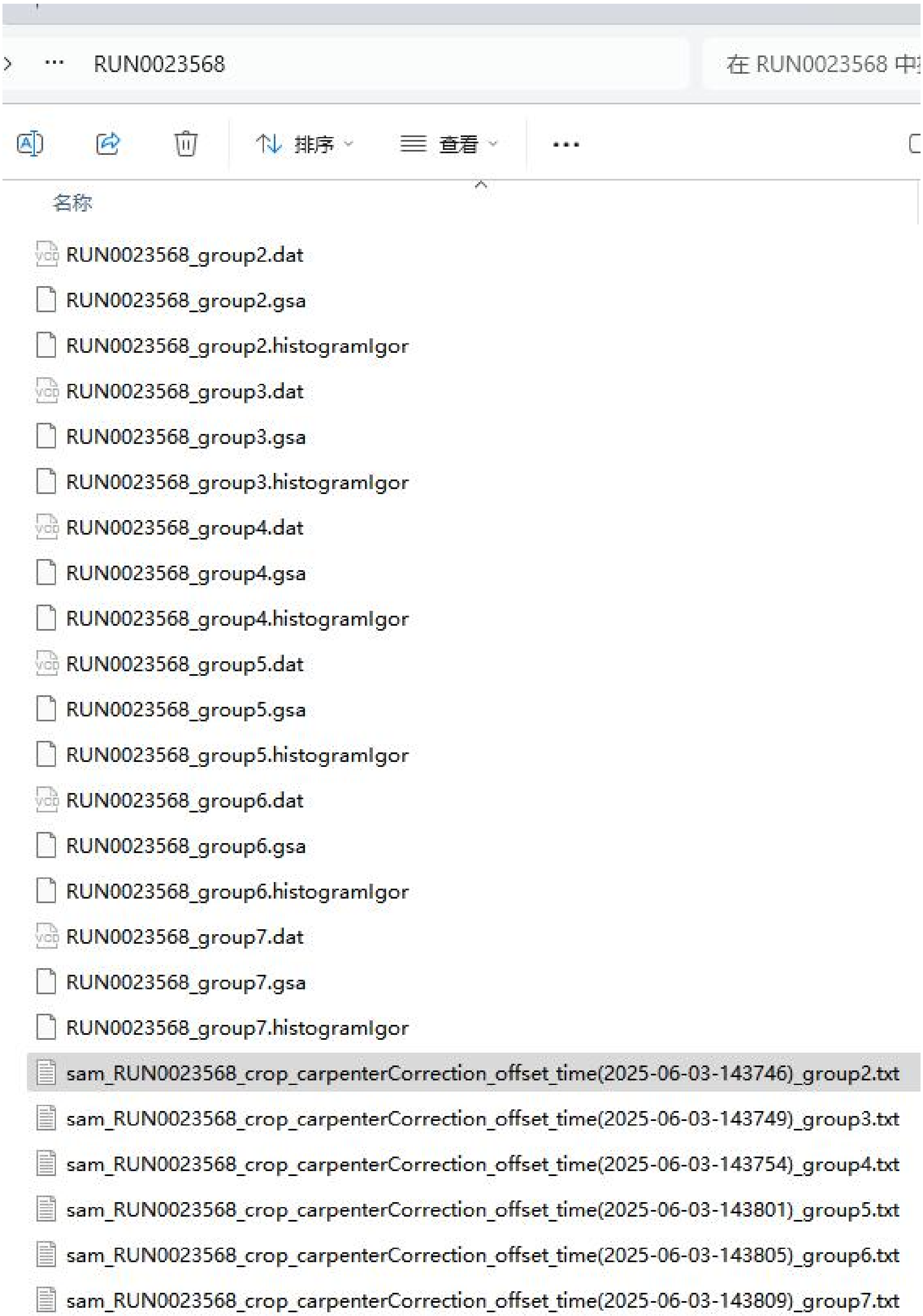


Figure 1.2-24 The saved reduced data. For each dataset, four files are saved: time-of-flight data files for data processing in the three software packages "FullProf", "GSAS", and "Z-Retiveld", as well as a txt file containing the d-spacing data.

1.2.8 Quick Reduction

After obtaining the time-focused data of V/VNi and its cell (container) through Sections 1.2.3 and 1.2.4, quick data reduction can be performed on the Quick Reduction page in RZera. The Figure 1.2-25 shows the interface of Quick Reduction along with configuration and usage instructions.

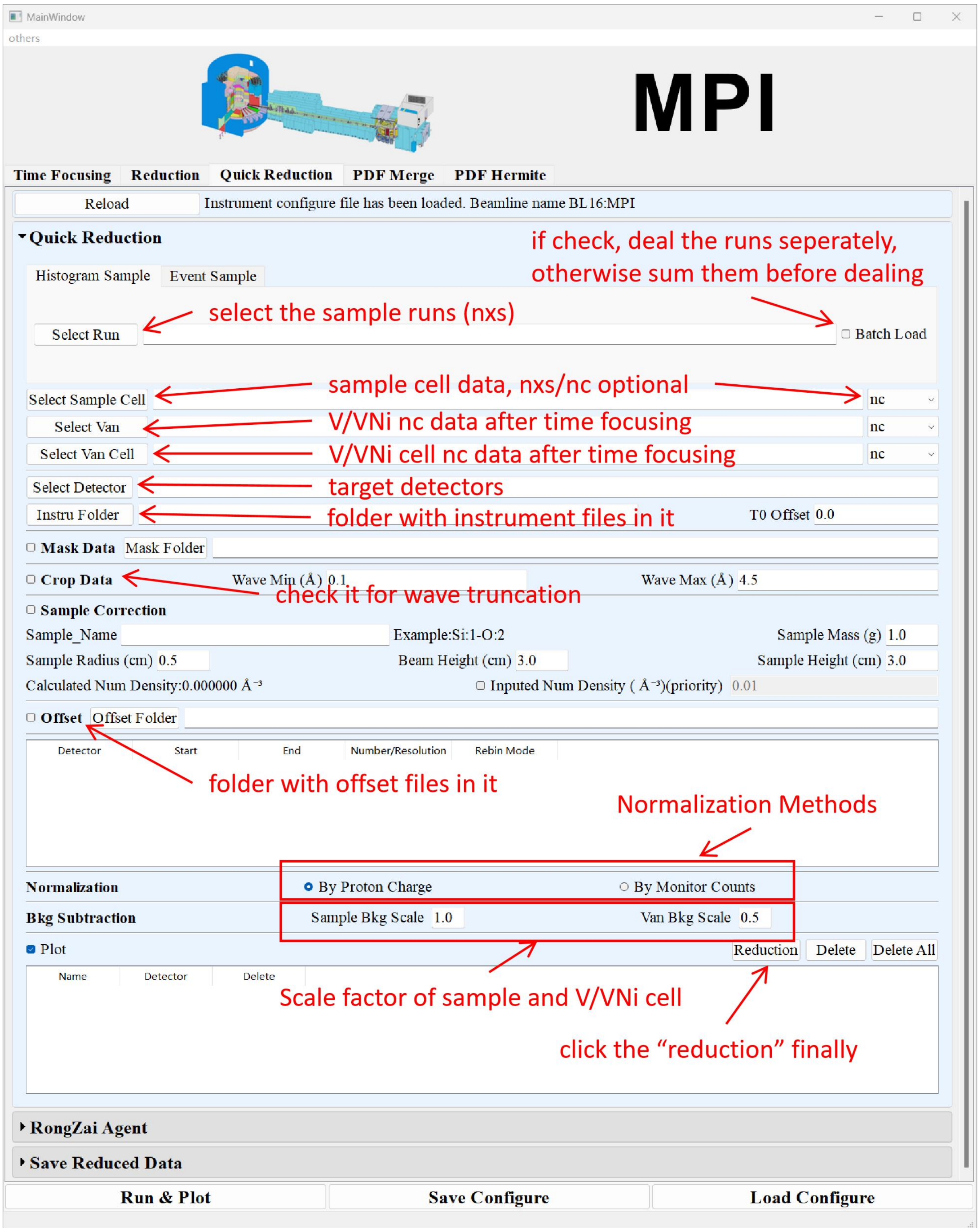


Figure 1.2-25 the interface and operation instruction of "Quick Reduction".

1.3 Using case data for data reduction.

The biggest difference between event data and histogram data is that event data retains the timestamp for each neutron count. Therefore, event data can be subjected to time slicing to observe diffraction patterns collected during different time intervals within the same RUN. This feature is particularly convenient for in situ experiments.

1.3.1 To obtain event data locally.

On the Time Focusing page, the second module is the "Get Event Data" module. This module allows users to transfer event data of specified RUNs (multiple RUNs allowed) to the server where RZera is hosted, for subsequent data reduction. The transfer method is illustrated in the Figure 1.3-1:

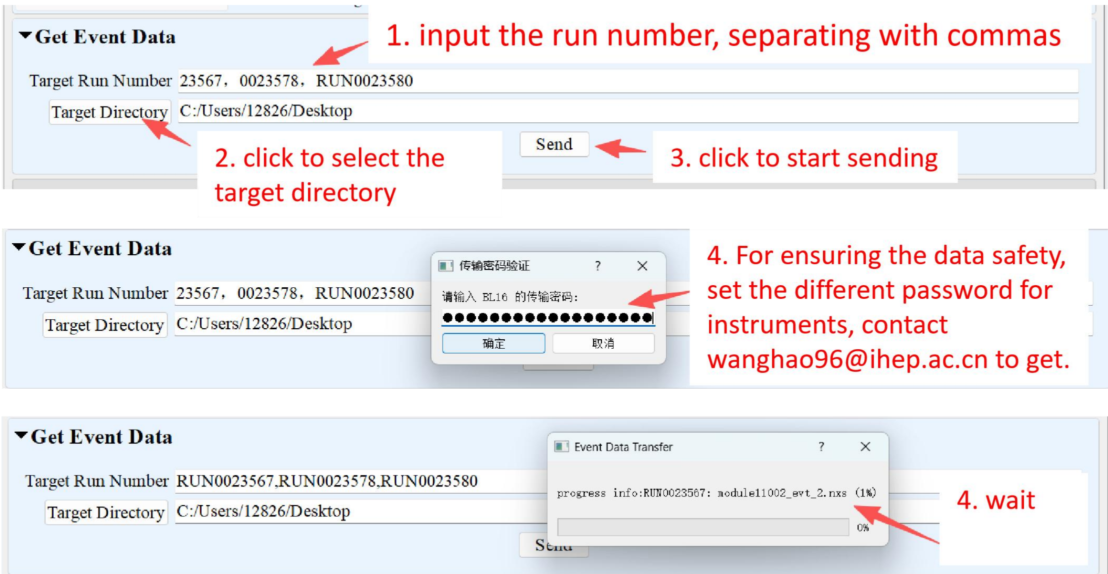


Figure 1.3-1 Schematic diagram of the Get Event Data module usage.

1.3.2 Reduction using event data.

Compared with the reduction using histogram data described in Section 1, the workflow is essentially the same, but the following two differences should be noted:

（1） The required preparation files are different.

Among the four types of files that need to be prepared—sample, V/VNi, sample background, and V/VNi background—only the sample no longer requires histogram data; instead, event data is needed. The other types remain exactly the same as before.

The histogram data file for the sample is shown in Figure 1.3-2. Under the RUN number folder, the mandatory files are the event data files for each module, named "moduleevt.nxs", along with the "pcLog.nxs" file. Moreover, it must be ensured that the number of event data files is the same for every module. In Figure 25, each module has one event data file. For other RUNs, there may be cases where each module has multiple event data files; in any case, please ensure that the number of files is consistent across different modules.

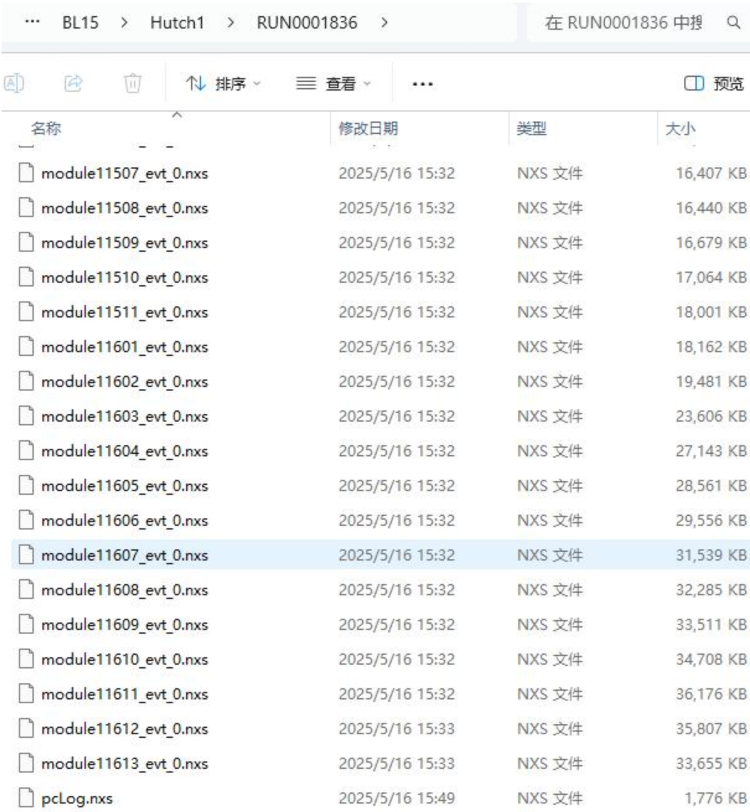


Figure 1.3-2 Event data collected from experiment RUN0001836 on the high-pressure spectrometer.

（2）The method of importing data in the "Time Focusing" page is different.

The interface for importing event data on the "Time Focusing" page is shown in Figure 1.3-3. The data is still imported in the "Load and Time Focusing" module. First, you need to switch to the "Event Data" tab within this module, then click the "Select Event Data" button and select the folder where the event data is stored, as shown in Figure 1.3-3(a). Please note that currently, event data only supports importing data from a single RUN; batch import and slicing of multiple RUNs are not supported. After the data is imported, the software will automatically read the total measurement time of that RUN and display it on the interface (Exp Time), as shown in Figure 1.3-3(b). Users can set the time bin width (default 8 μs) for the histogram data obtained after time slicing, and most importantly, they can configure the time slicing parameters. As shown in Figure 1.3-3(b), the time slicing parameters are presented as a three-column table, with rows that can be added or removed using the "Add" and "Remove" buttons. Each row consists of three parameters: slice start time, slice end time, and the number of equal intervals to divide this time range into. The slicing parameters entered in Figure 1.3-3(b) mean that one segment is cut from 0–10 min, and two segments are cut from 10–20 min. This will result in three data segments: 0–10, 10–15, and 15–20 min.

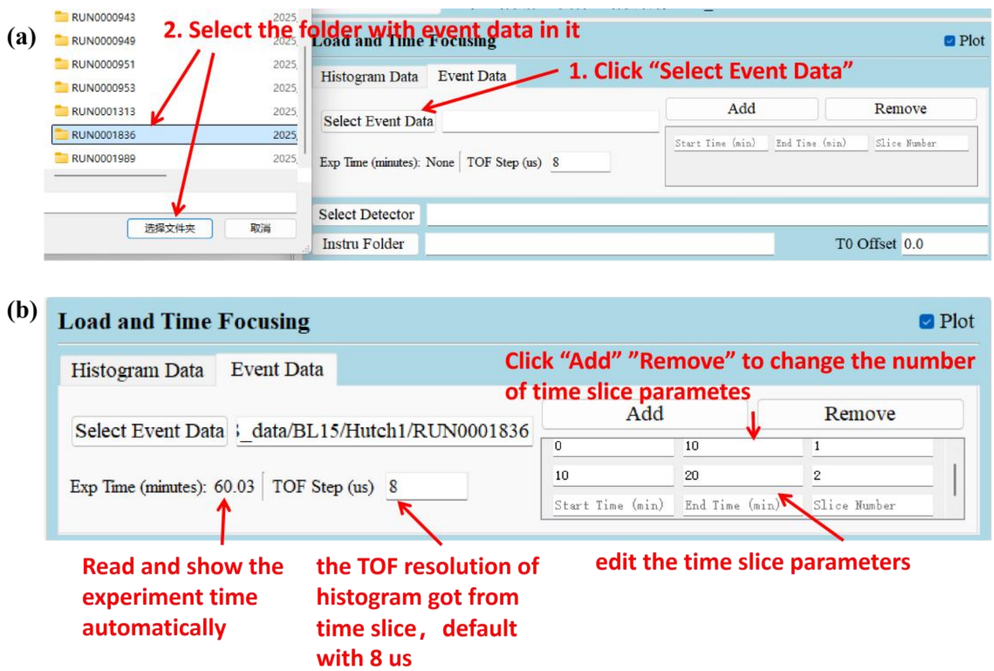


Figure 1.3-3. Instructions for importing event data in "Load and Time Focusing".

1.4 Obtaining PDF (Merge/Hermite) from time-focused data.

If users need to obtain PDF data, they must first follow the steps in Section 1 to obtain the time-focused data files for the four types of files, and then proceed according to the instructions in this section. RZera provides two methods for obtaining PDF data: one is to obtain PDF data after merging the diffraction patterns, and the other is to obtain PDF data through Hermite polynomial fitting. The procedures for each method are described below:

1.4.1 Obtaining PDF data after merging.

Step 1: Switch to the "PDF Merge" page and perform data reduction.

Users first need to switch to the "PDF Merge" page, and will notice that the initial several modules on this page are exactly the same as those on the "Reduction" page. This is because the data must be reduced before obtaining PDF data. Up to the "Division" module, the operation procedure is identical to that described in Section 1, and thus will not be reiterated here.

Step 2: Convert the reduced I(d) data to S(Q) data.

Next, the user needs to operate the subsequent "Convert to S(Q)" module to convert the I(d) data into S(Q) data, as shown in Figure 1.4-1.

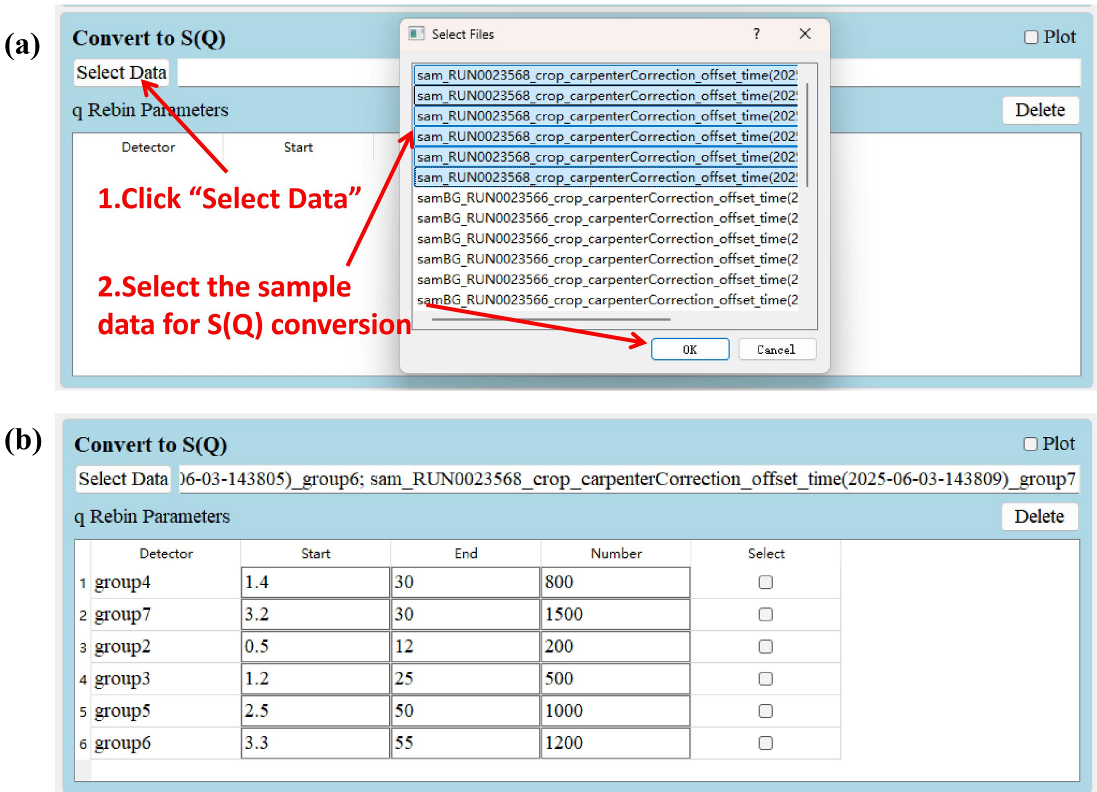


Figure 1.4-1 (a) Procedure for selecting the sample data to be converted to S(Q). (b) After the selection is completed, the default q Rebin parameters for the detectors associated with the selected data will automatically appear on the interface. These parameters are modifiable, and users can adjust them according to the actual data conditions.

After processing the data as instructed in Figure 1.4-1, users are strongly advised to evaluate the S(Q) data before proceeding further. To evaluate the data, first check the "Plot" option in the "Convert to S(Q)" module, then click "Run & Plot" at the bottom of the page. The software will execute the modules one by one and generate plots for those with the plotting option enabled. The plotting results are shown in Figure 1.4-2. The figure indicates that data from group 2 and group 4 are problematic. After adjusting the q rebin parameters, the issues were improved, as shown in Figure 1.4-3.

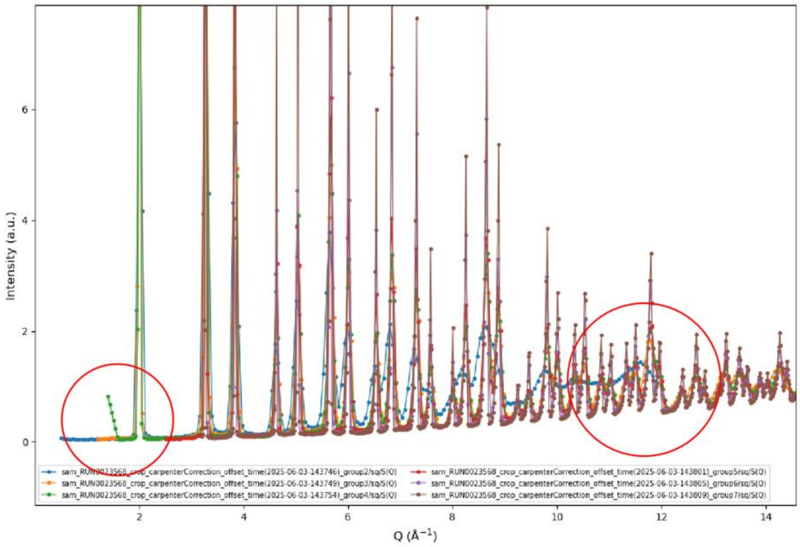


Figure 1.4-2. Plotting results of S(Q) data converted using default parameters. A normal S(Q) dataset should approach 1 at high Q. It can be seen that the blue curve (group 2) is clearly problematic. In addition, the green curve (group 4) exhibits an unnatural upturn at low Q, which also needs to be addressed.

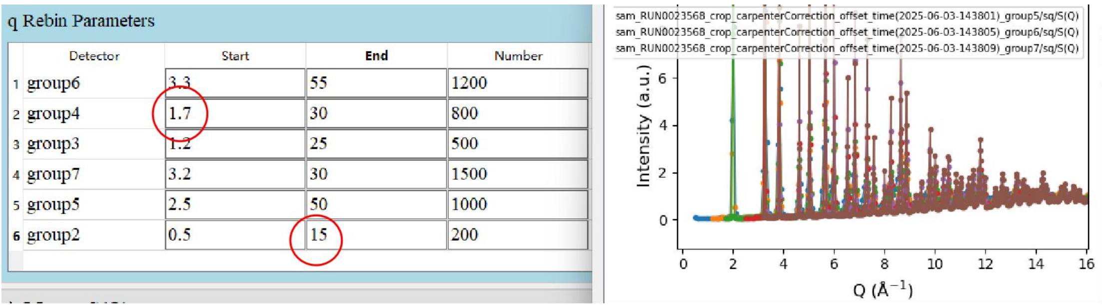


Figure 1.4-3. The q rebin start parameter for group 4 was increased to truncate the upturn portion. The q rebin end parameter for group 2 was raised to include the previously truncated valid data, and the convergence of the curve was improved.

Step 3: Merge the S(Q) data into a single dataset.

As can be seen in the previous step, the Q-range coverage differs significantly among detectors from different groups. For PDF analysis, the low-Q region is particularly important; however, the dataset from group 2, which has the lowest Q 下限, covers only a very limited range at high Q. Therefore, it is necessary to merge the data from different coverage regions into a single dataset for subsequent PDF generation.

First, expand the "Merge S(Q)" module, then click "Load Base Data" to select the base dataset for merging. Next, click the "Add" button to select other data to be merged with the base data. At this point, a row will appear in the list, with two parameters to fill in: "overlap start" and "overlap end". These two parameters determine the position of the merging node when combining the two datasets. Before the data segment defined by these two parameters, the base data is used; after it, the other data to be merged is used, and this segment is processed to ensure a smooth connection between the two datasets. It can be seen that the merged data is a concatenation of the two segments. After that, you can continue to click "Add" to include more datasets, and the module will merge the data sequentially from top to bottom. Finally, based on the Q rebin parameters set in this module, a new merged curve is generated. The entire workflow is shown in Figure 1.4-4.

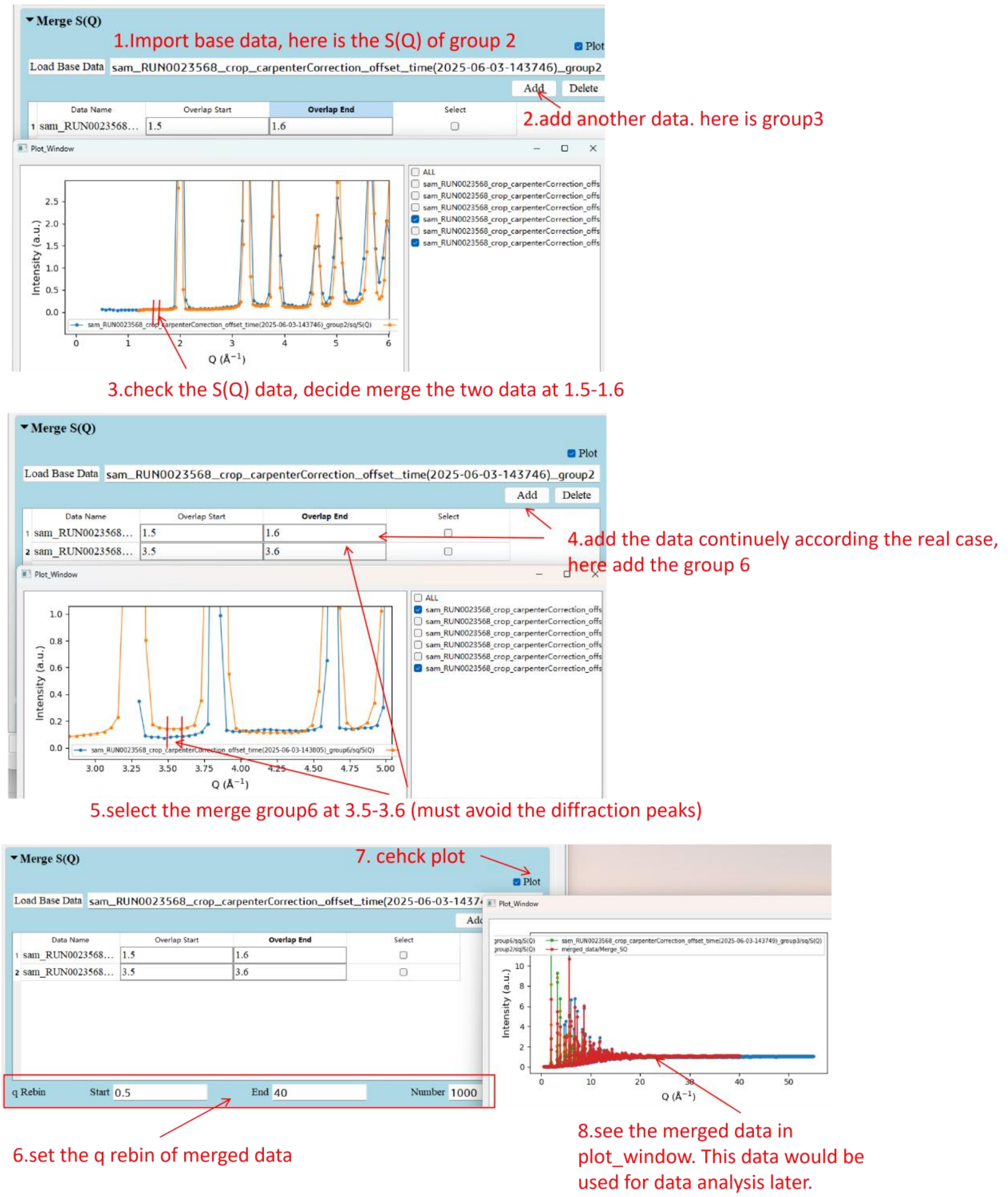


Figure 1.4-4. Schematic diagram of the workflow for using the Merge S(Q) module.

Step 4. Save S(Q) (Optional)

Immediately after the Merge S(Q) module, there is a "Save S(Q)" module. This module is used to save the S(Q) data. If needed, simply expand it and specify a save path; if not, you can just collapse it.

Step 5. Configure the "Get PDF" module to obtain PDF data.

As shown in Figure 1.4-5(a), expand the "Get PDF" module, then configure the r Rebin parameters. These three parameters determine the start, end, and number of data points of the resulting PDF data. The "Lorch" option is generally recommended to be checked, because using a limited S(Q) range for PDF conversion introduces truncation effects that cause periodic oscillation artifacts in the PDF data; this option is used to suppress such oscillations. The "Self Term" is used in calculating Qi(Q) = Q*(S(Q) - Self Term), with a default value of 1, which usually does not need to be changed. There are three PDF types available: "G(r)", "g(r)", and "RDF". Note that the naming convention for PDF data can be somewhat confusing. Here, G(r) is also referred to as D(r) in some contexts, and it is a commonly used data type (it should be noted that under the nomenclature where D(r) exists, G(r) also exists, but that G(r) is different from the one in this software. In this software, G(r) is equivalent to D(r) under that naming system; please be aware of this distinction). After checking "Plot", click "Run & Plot", and the PDF data will appear in the plotting window, as shown in Figure 1.4-5(b).

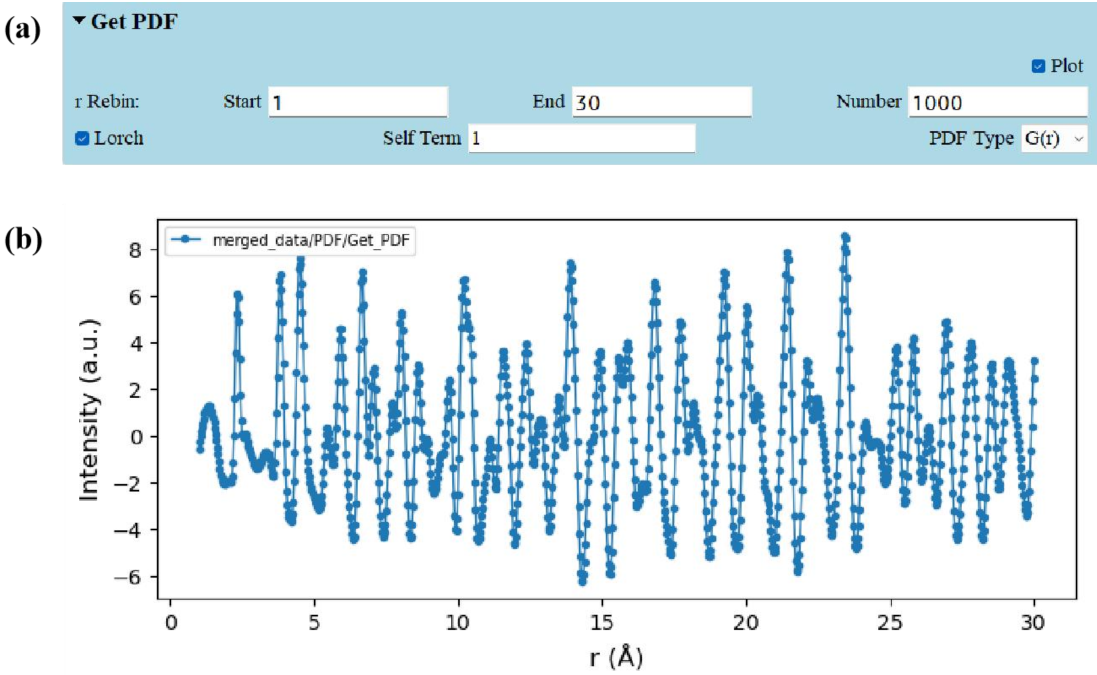


Figure 1.4-5 (a) The "Get PDF" interface after configuration. Note that the r Rebin parameters can be adjusted according to actual conditions, while the "Self Term" generally does not need to be modified. (b) The plotted PDF data "G(r)".

Step 6. Configure the "Save PDF" module to save the PDF data.

Expand the final "Save PDF" module and specify a local save path. As long as this module is configured and kept expanded, clicking "Run & Plot" will save the data locally. The saved data is shown in Figure 1.4-6.

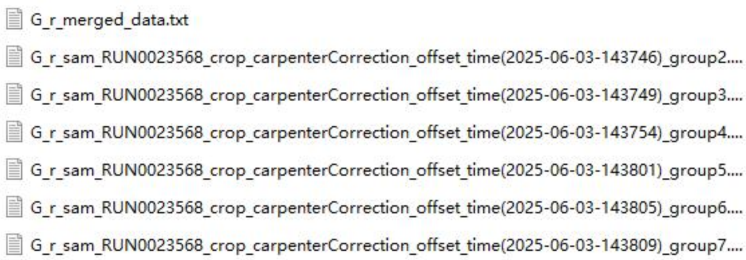


Figure 1.4-6. The saved PDF data, including the PDF from the merged data as well as the PDFs from individual detectors before merging. In short, as long as S(Q) data are generated during the processing, they will all be converted to PDF. Among them, the user may only need the PDF from the merged data, so selection should be made according to actual needs.

1.4.2 Obtaining PDF data via Hermite polynomial fitting.

Using the merge method to obtain PDF data has a notable drawback: different detectors have different resolutions, and the counts of the same diffraction peak vary among detectors. Directly merging the data together essentially ignores the effects of these factors. RZera provides an alternative method for obtaining PDF data, namely through "Hermite polynomial fitting", which yields PDF data free from instrumental broadening effects. This method was developed by Martin T Dove, and with his assistance, it has been integrated into RZera for use by users at spallation neutron sources. The underlying theory is not discussed here; only the usage instructions are provided.

Step 1: Switch to the "PDF Hermite" page and perform data reduction and conversion to S(Q).

Users first need to switch to the "PDF Hermite" page, and will notice that the initial several modules on this page are still exactly the same as those on the "Reduction" page. This is because the data must be reduced before obtaining PDF data. Up to the "Division" module, the operation procedure is identical to that described in Section 1, and thus will not be reiterated here. After that, the subsequent "Convert to S(Q)" module is used in exactly the same way as described in the first part of this section ("Obtaining PDF Data via Merge"), so it will not be repeated here either.

Step 2: Configure the Hermite Fitting module.

Expand the Hermite Fitting module and fill in the required fields, as shown in Figure 1.4-7. First, the q Rebin parameters need to be specified. Note that the q Rebin here should cover the range of all S(Q) data that are to be included in the fitting. For example, if the S(Q) data from all detectors range from 0.5 to 55 and I wish to use all of them in the fitting, then the Min and Max in q Rebin should be set to 0.5 and 55. If I set them to 0.5 and 50, then the data above 50 from group 7 will not participate in the fitting. The r Rebin parameters can be set according to one's own needs. In the "Get and Preprocess qI(q)" subsection of this module, the "Order of Hermites" can generally be left at the default "Auto". The "Self Term" can remain at the default value of 1. "Chebyshev Fitting" should normally be checked. The "Fitting Order" defaults to 8 and generally does not need to be modified; if modified, it must be set to an even number. If the sample is crystalline, "Crystal" should be checked; if it is amorphous, it does not need to be checked. "Lorch" should normally be checked. The "Plot" option here is not for plotting PDF data, but for plotting the preprocessed qI(q) data. Before performing PDF fitting, users can check whether the qI(q) data appear normal. In this case, the user can leave "Hermite Fitting" unchecked below, then click "Run & Plot" to view qI(q), as shown in Figure 1.4-8.

After confirming that the Qi(Q) data are normal, check the "Hermite Fitting" option. If "Fitting with Instrument Resolution" is also checked, the instrument parameters will be convolved during the fitting process, resulting in PDF data that are decoupled from the instrumental resolution.

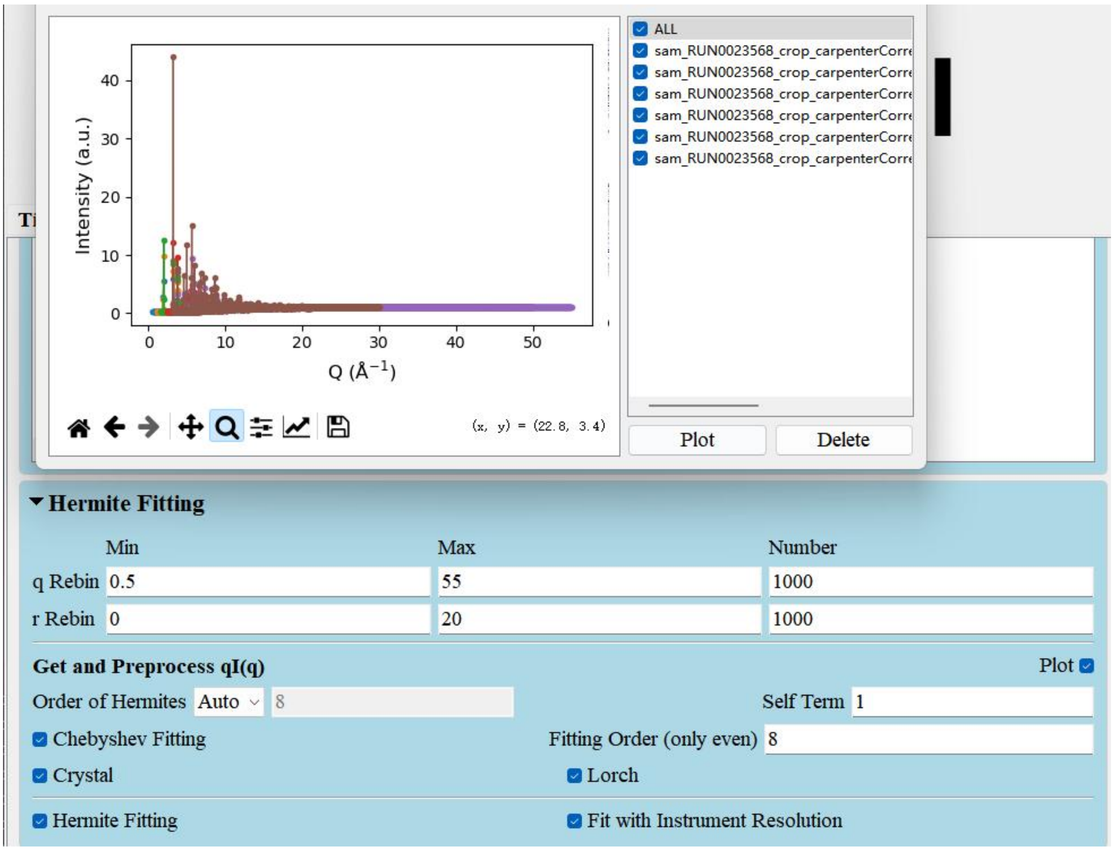


Figure 1.4-7. The configured Hermite Fitting module.

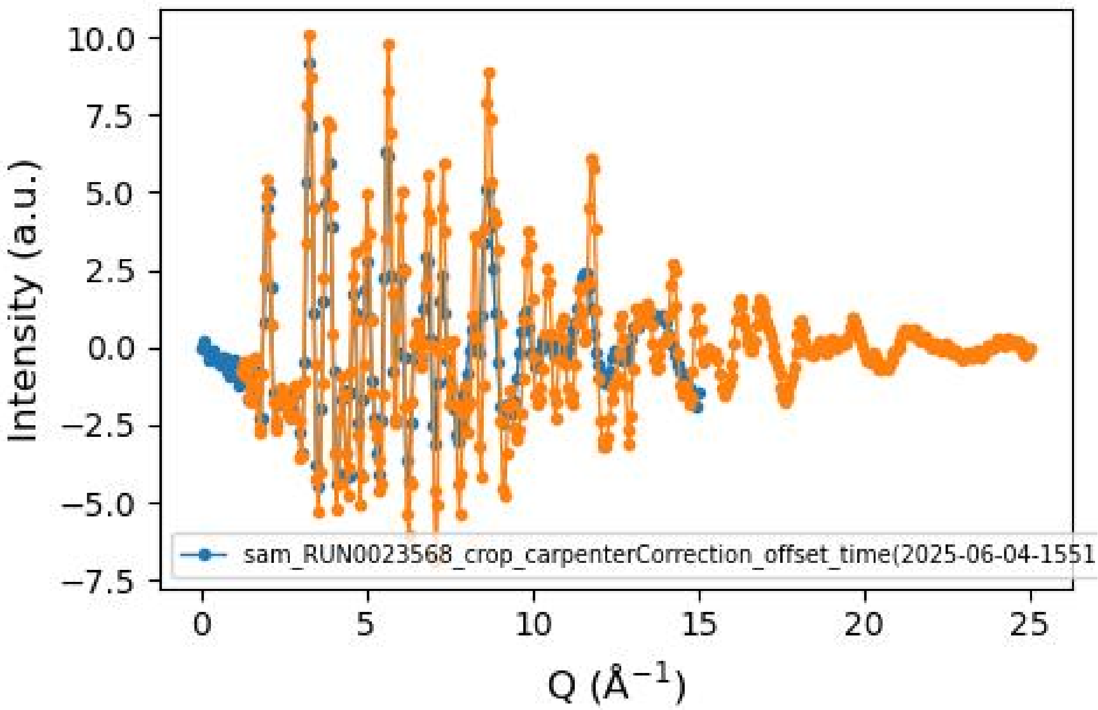


Figure 1.4-8. The processed qI(q) data. The key points in evaluating this data are whether it tends toward 0 at low Q and whether it also tends toward 0 at high Q.

Step 3: Configure the final Save PDF module.

At the end of this page, there is a "Save PDF" module, as shown in Figure 1.4-9(a). In this module, the "Plot" option and the PDF type can be set. If "Plot" is checked, clicking "Run & Plot" will plot the PDF result obtained from the Hermite fitting, as shown in Figure 1.4-9(c). The plotted PDF type corresponds to the one specified in this module. Meanwhile, if a "Save Path" is specified, the module will save the PDF data to the designated path, as shown in Figure 1.4-9(b).

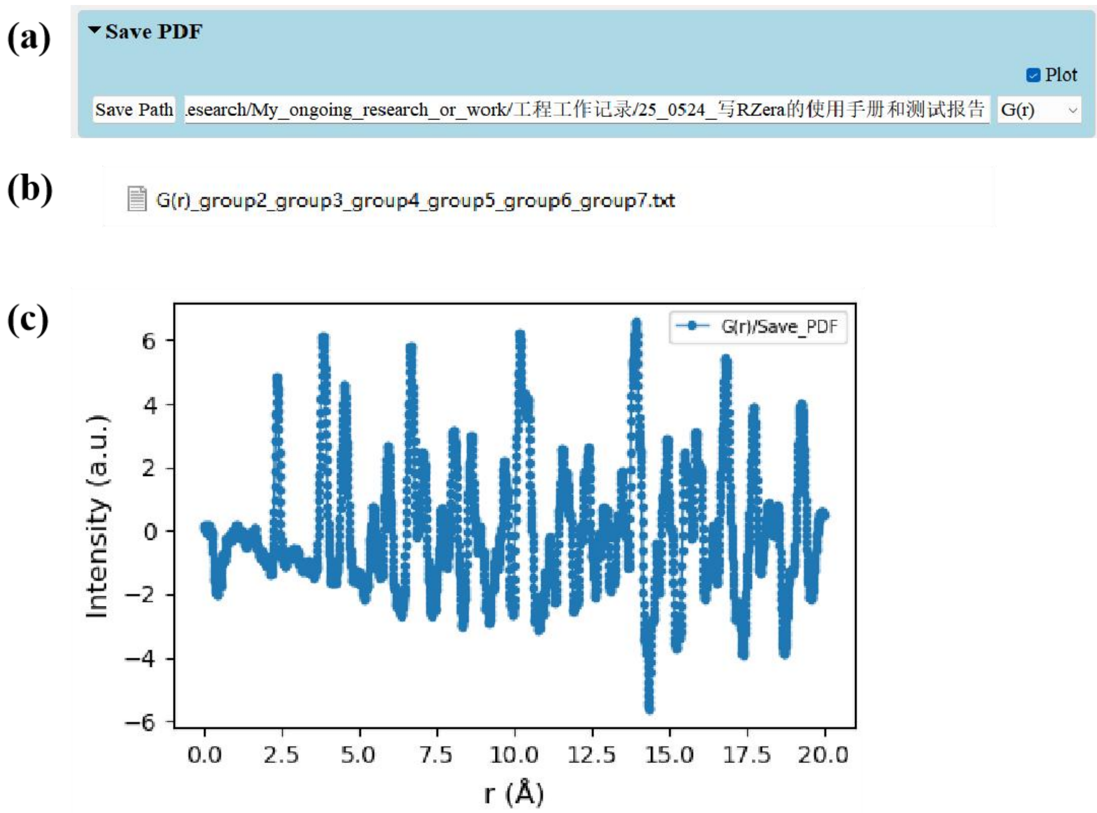


Figure 1.4-9 (a) The configured "Save PDF" module. (b) The saved PDF data. (c) The plotted PDF data G(r).

Step 4 (Optional): Configure the "Get S(Q) from PDF" module and the "Save S(Q)" module.

The purpose of these two modules is to inversely calculate the corresponding S(Q) data from the final fitted PDF data. These two modules are provided because when obtaining PDF via Hermite fitting, the process starts directly from the S(Q) data of each detector without merging, and thus does not produce a corresponding merged S(Q) dataset. Considering that such data may be needed for subsequent analysis, these two modules have been added.

As shown in Figure 1.4-10(a), simply configure the q Rebin and Save Path. It is recommended that the q Rebin here be kept consistent with that in the "Hermite Fitting" module. The calculated S(Q) data are shown in Figure 1.4-10(b). The saved data are shown in Figure 1.4-10(c).

。

(a)

(b)

(c)

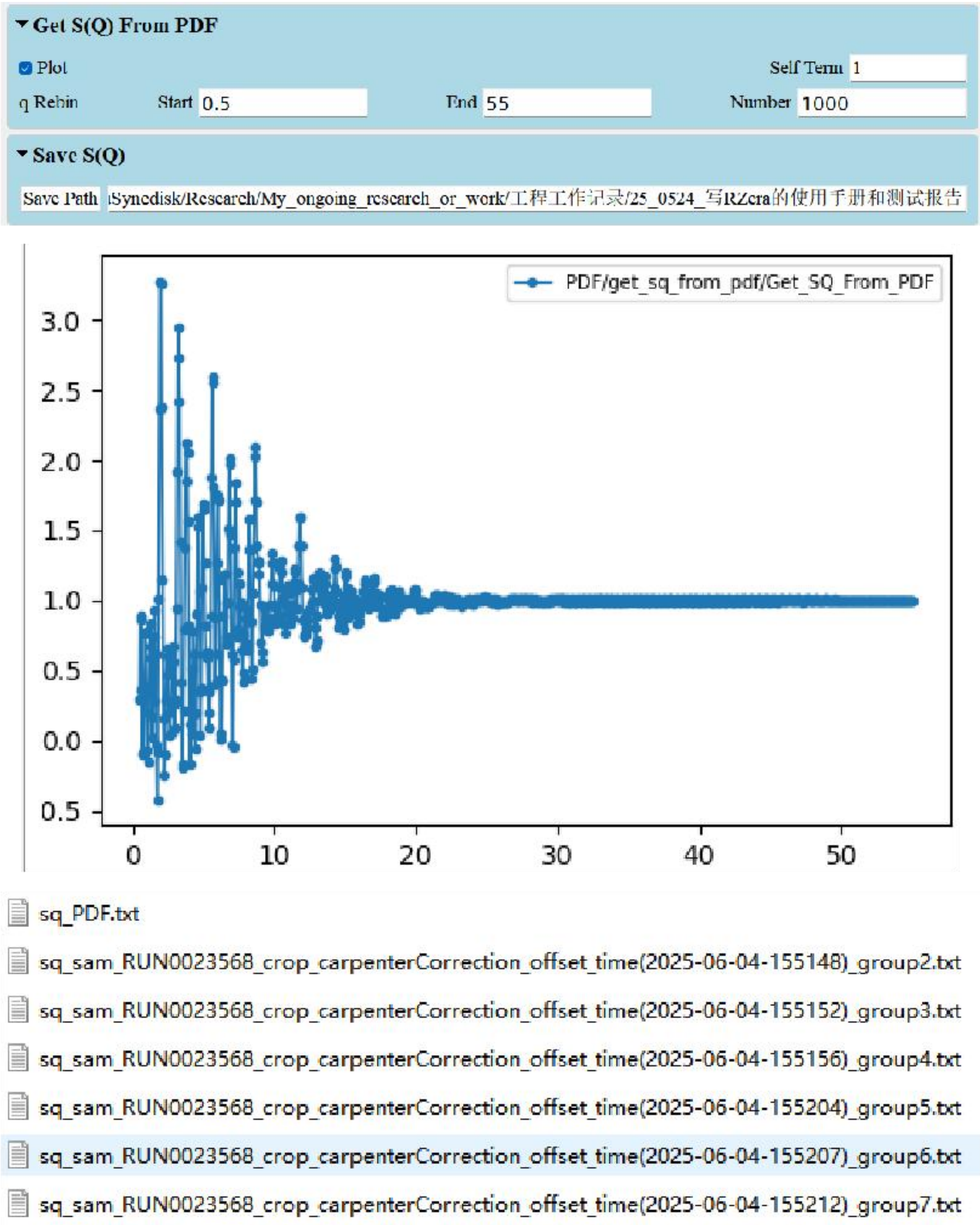


Figure 1.4-10 (a) The configured "Get S(Q) from PDF" module and "Save S(Q)" module. (b) The calculated S(Q) data. It can be seen that a notable difference from the real S(Q) data is the presence of obvious periodic oscillations, which must be noted. (c) The saved data. Among them, "sq_PDF.txt" is the calculated S(Q) data, while the others are the real S(Q) data from different detectors, all of which are saved together.

1.5 Quick guide for obtaining the offset file (currently only supports BL16).

The offset file is commonly used when performing Time Focusing on data, and it is typically provided by the instrument. This section describes how to obtain it when no offset file is available. Acquiring an offset file involves adjusting many parameters and is a rather complex process; however, RZera provides configuration files that already contain well-tuned parameters, so users only need to import the configuration file. The procedure is described below:

1.5.1:Open the offset interface.

In the upper-left corner of the reduction interface, there is an "Other" option. Click it, and from the dropdown menu, select "offset_correction". After clicking, you will see the interface shown in Figure 1.5-1.

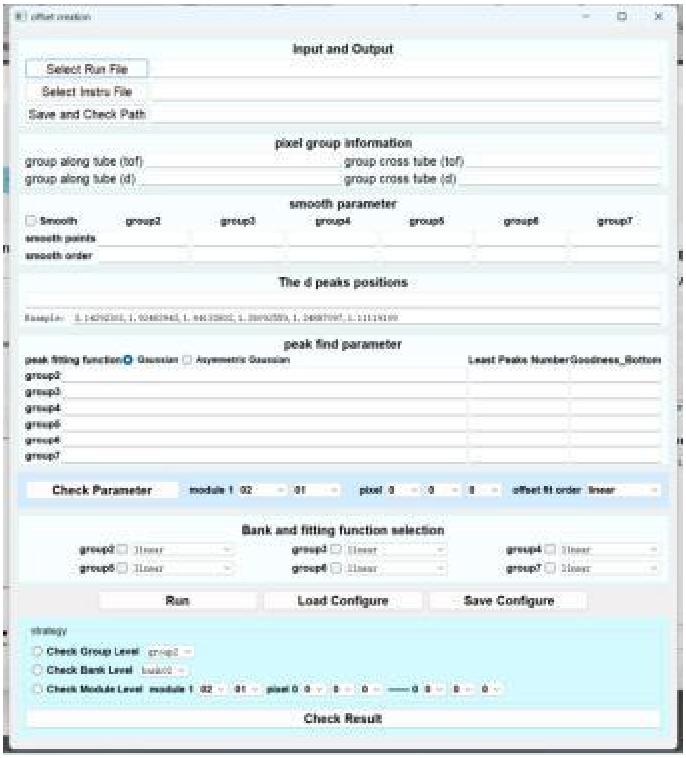


Figure 1.5-1. The operation interface for obtaining the offset file.

1.5.2:Import the configuration file.

As shown in Figure 1.5-2, click "Load Configure" and select the configuration file provided by RZera to import it.

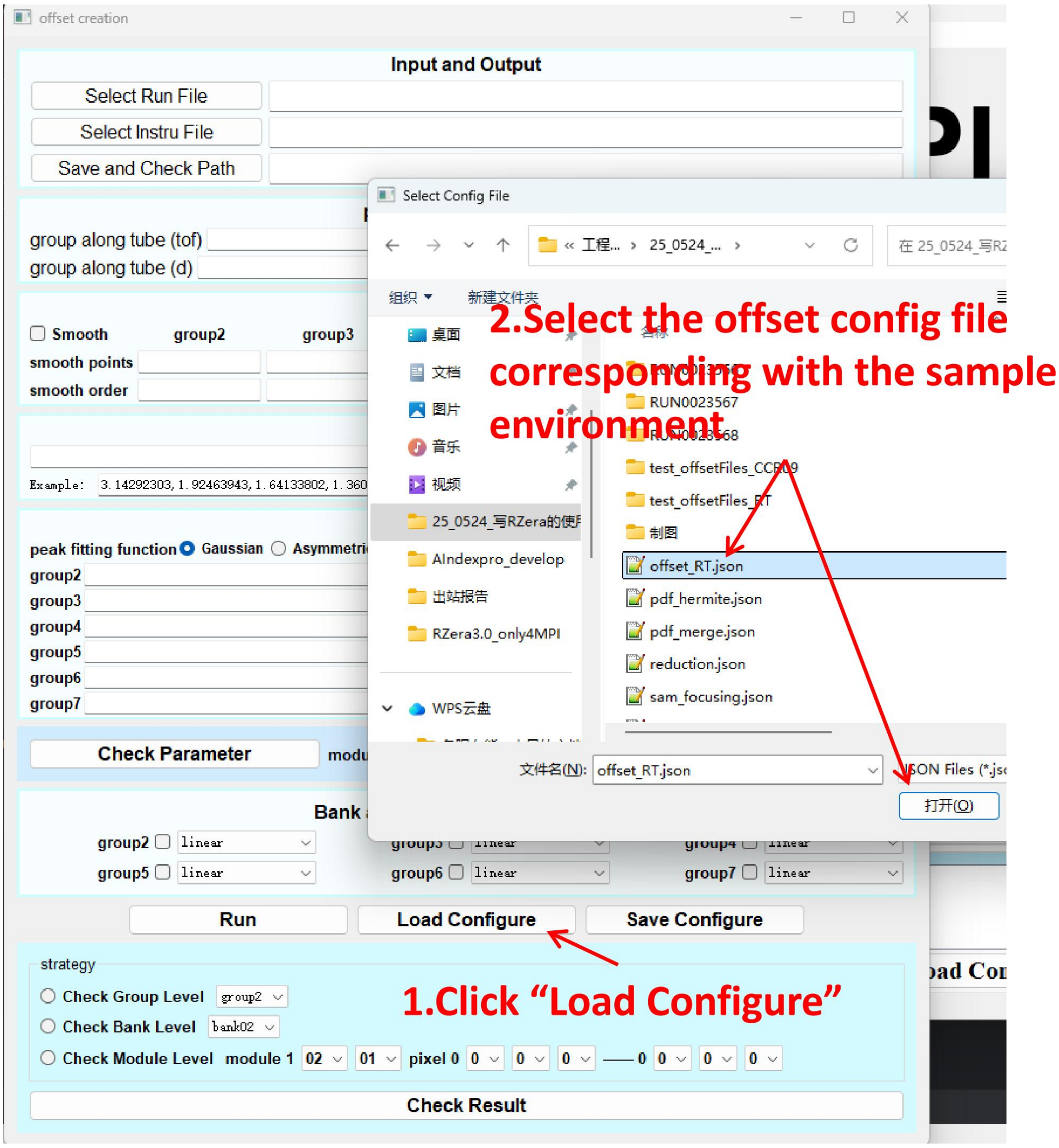


Figure 1.5-2. Method for importing the configuration file.

1.5.3:Modify the three directory paths.

At the top of the interface, there are three paths. The first is the path to the Si standard sample used for offset correction (note that it must be Si, not any other sample; otherwise, other parameters will need to be modified). The second is the path to the pidinfo folder (this folder is already provided with the software). The third is the folder path where the generated offset file will be saved. Users need to ensure that these three paths are correct, as shown in Figure 1.5-3. After modifying the paths, users can click "save configure" to save the JSON file containing the correct paths for future use, eliminating the need to modify the paths again.

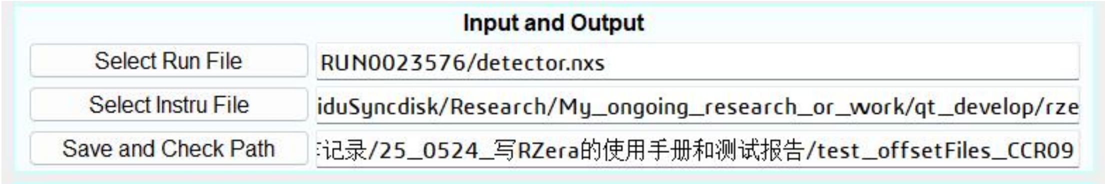


Figure 1.5-3. The three paths that need to be checked. Click "Select Run File" to import the Si experimental data to be used as the offset reference. Click "Select Instru File" to import the instrument pixel position file. Click "Save and Check Path" to specify the storage path for the offset file.

1.5.4:Click the "Run" button and wait for the file acquisition to complete.

After clicking "RUN", a progress bar will appear, as shown in Figure 1.5-4. Once the progress bar completes, the offset files have been successfully generated. Users can then retrieve these files from the path they specified in the previous step for saving the offset files.

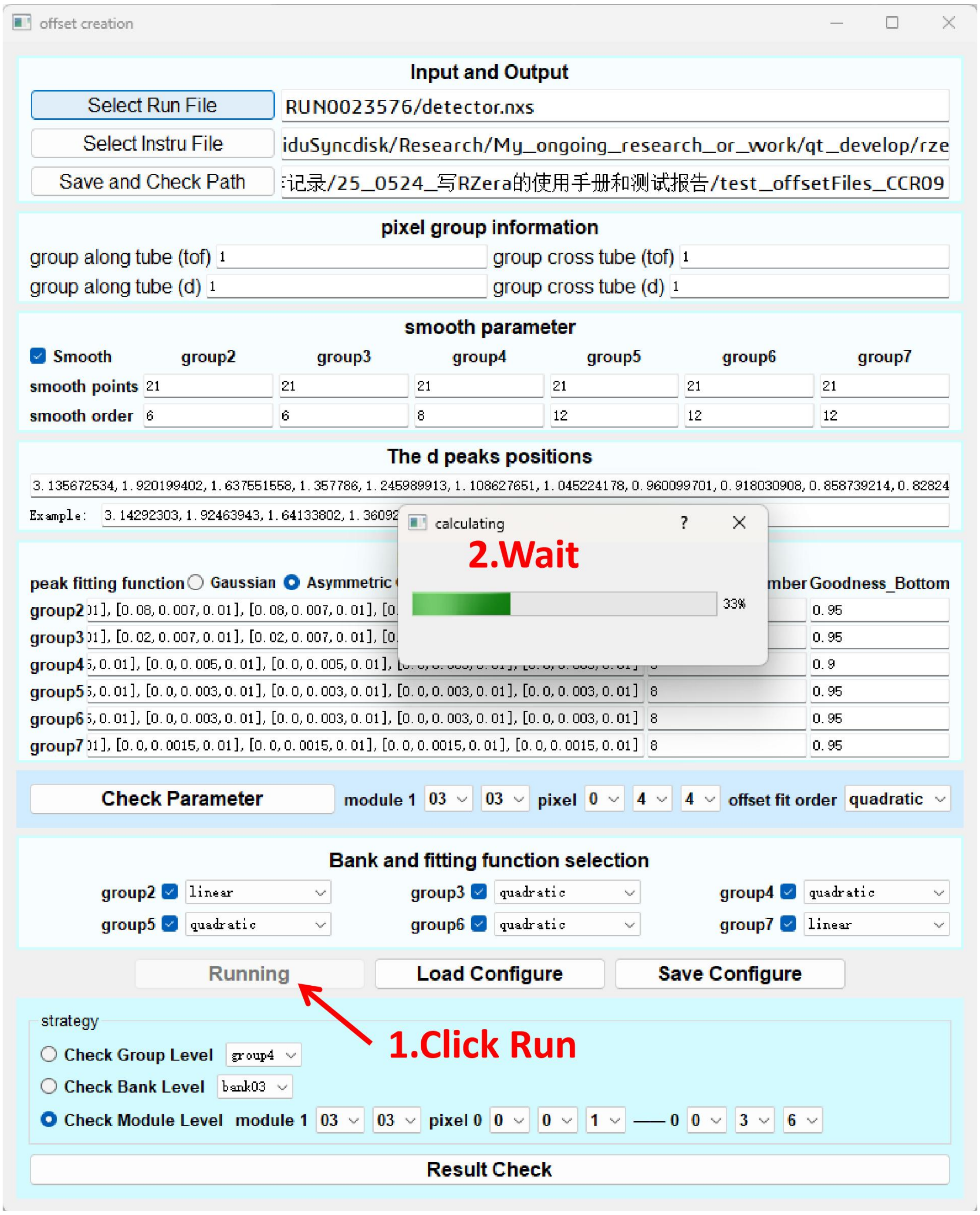
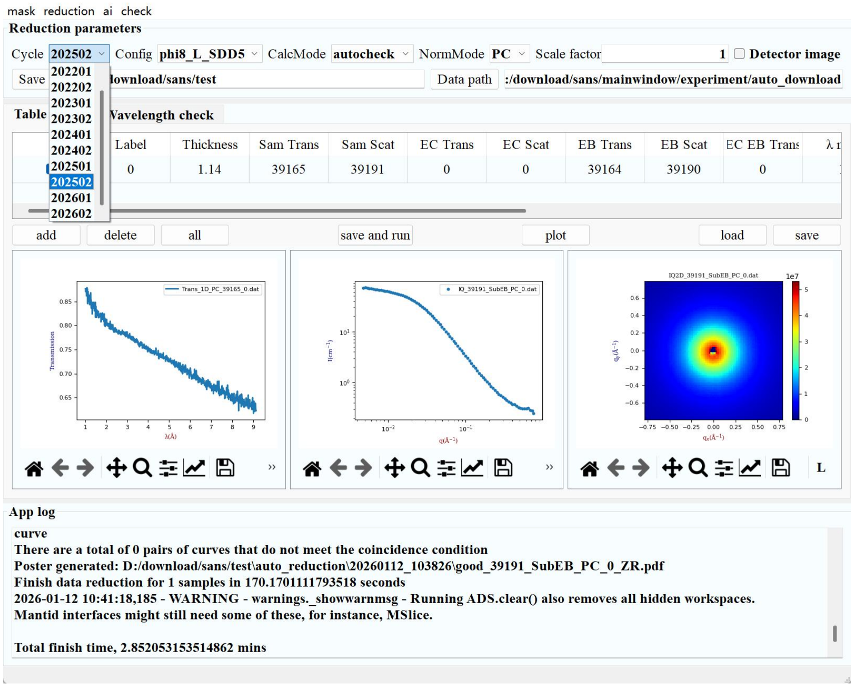


Figure 1.5-4. Waiting for the offset file to be generated after clicking Run.

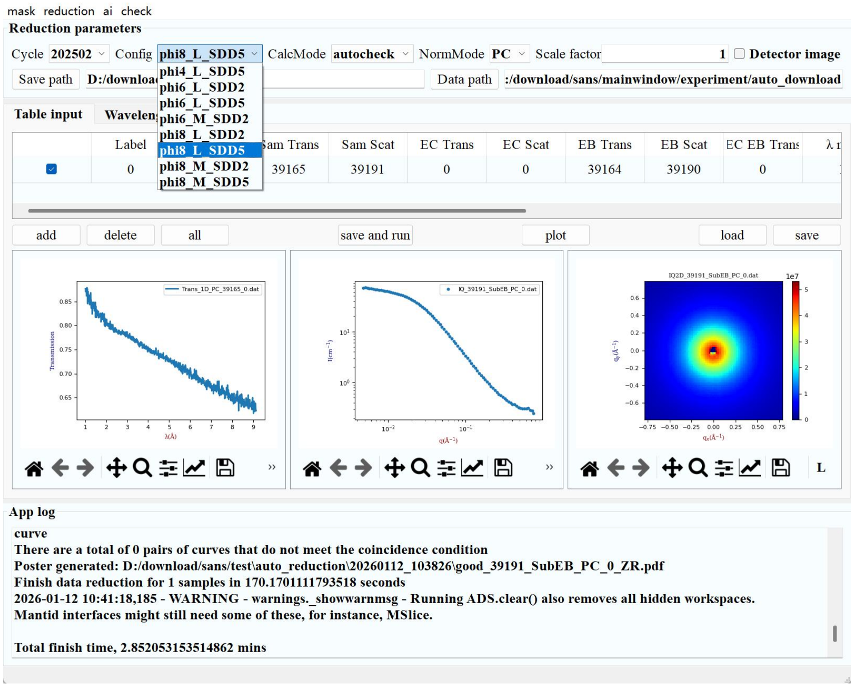
2. Operating Guide for the Small-Angle Scattering Spectrometer Software

2.1 Introduction to the reduction interface functions.

1、**Select Cycle:** The instrument parameters are selected by year, with each year divided into two halves, represented as 01 and 02 in the options. For example, 202501 indicates that the instrument parameters are those calibrated in the first half of 2025.

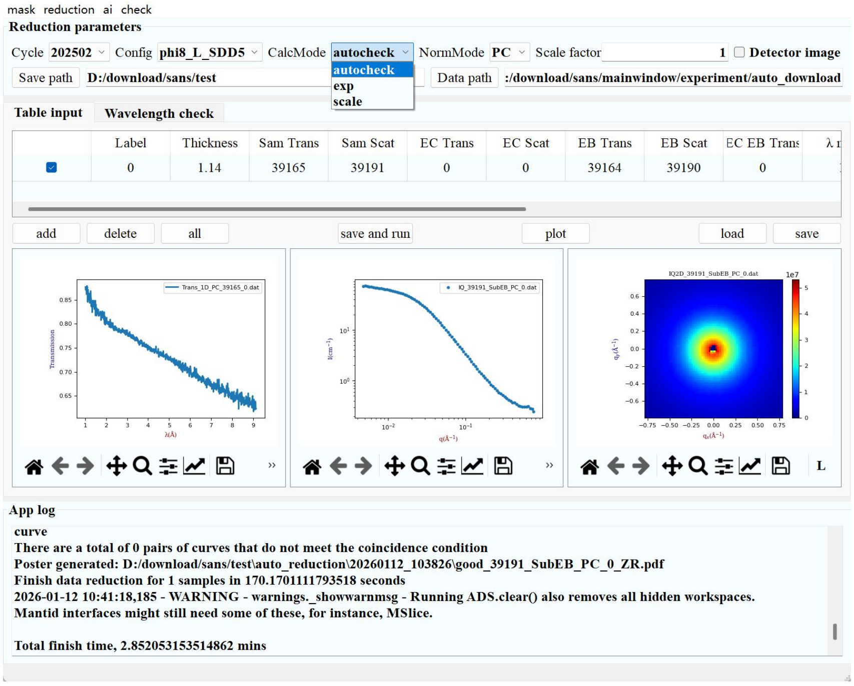


2、**Select Config:** This parameter determines the instrument calibration folder used by the software. Please select the correct detector configuration.



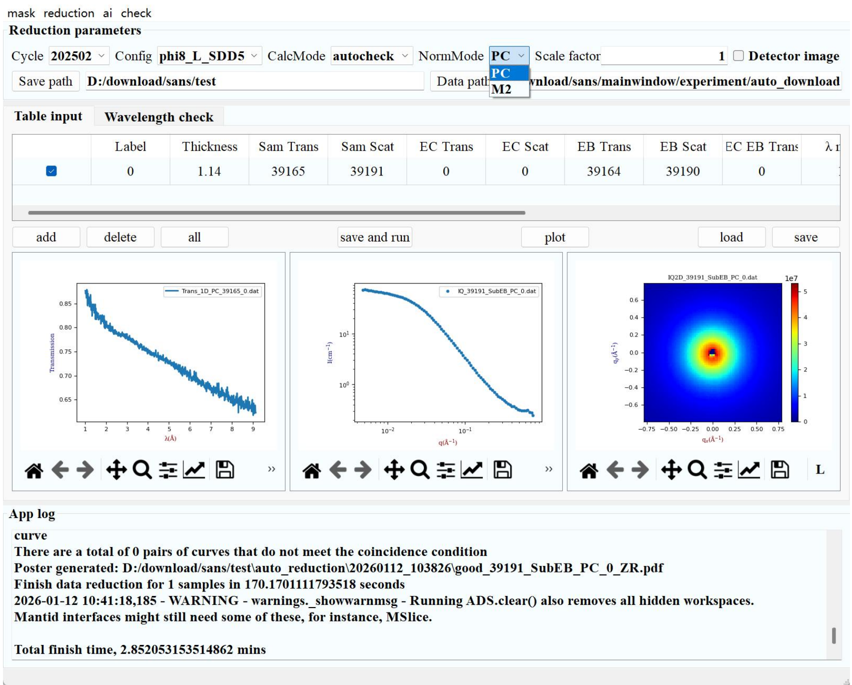
3、**Select CalcMode:** Calculation mode.

- Scale: Scale factor calculation mode. In this mode, only one experimental dataset can be selected.
- Exp: Reduction mode. In this mode, reduction is performed for all selected experiments.
- autocheck: Automatically completes Bragg edge judgment and orientation determination, and uses appropriate wavelength ranges and integration angles to perform automatic reduction. The judgment results are saved as a PDF file for manual verification, significantly improving reduction efficiency.
  This mode is recommended.



4、**Select NormMode:** Normalization mode selection.

- PC: Normalization using neutron beam intensity.
- M2: Normalization using monitor M2.



5、**Scale factor：** ref data / exp data, calculated from the scale mode.

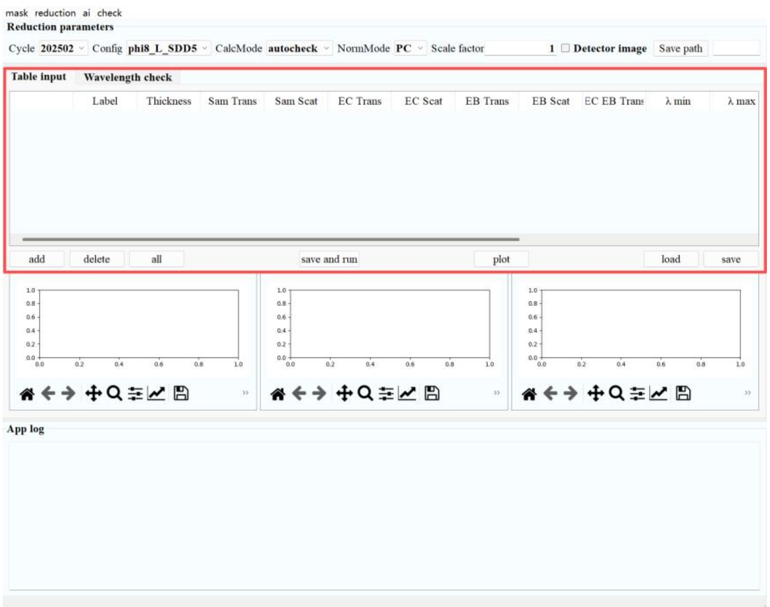
6、**Detector image：** Check this option to save the 2D detector images generated during the calculation process. Leave it unchecked to skip saving.

7、**Save path：** Save path for the calculation results.

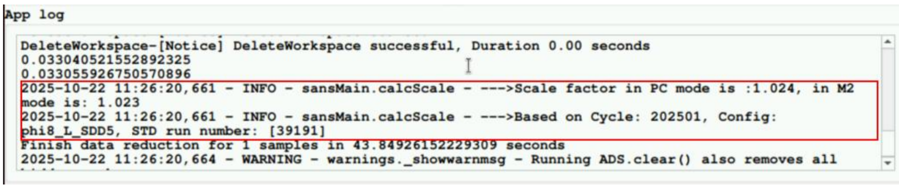
8、**Data path：** Path to the data files.

9、Fill in the experimental data.

- **load：** Import an Excel file containing sample thickness, experiment, can, background RUN numbers, and wavelength range.
- **save：** Save the table in Excel format.
- **add：** Add a blank row after the last row of the table.
- **delete：** Delete the checked rows.
- **all：** Check or uncheck all rows.
- **save and run：** Perform a calculation and save the generated files to the save path. The 1D, 2D, and Trans folders contain the plotting and analysis results. The corresponding files can be found based on the RUN numbers of Sam Scat and Sam Trans.
- **plot：** Plot the transmittance curves, 1D scattering curves (double log), and 2D scattering images for all checked experiments. Note: Only one 2D image is plotted; if multiple experiments are checked, only the image corresponding to the last checked experiment is displayed.

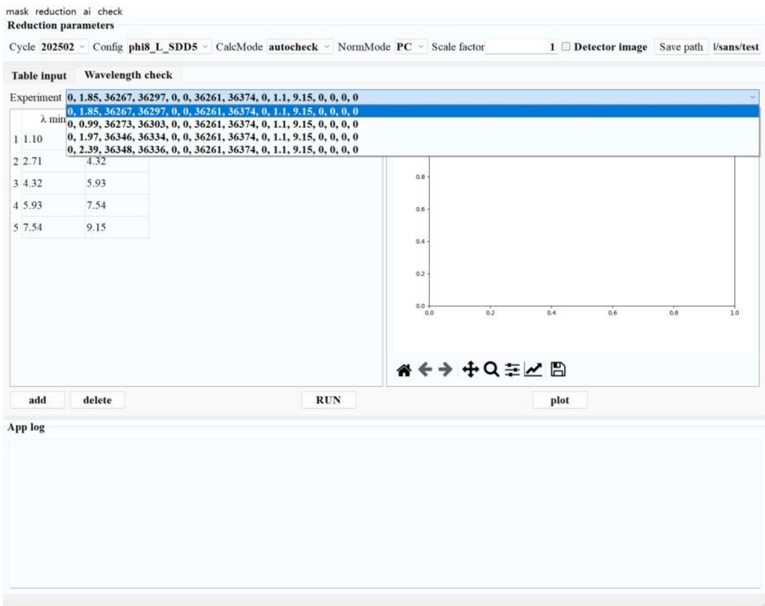


10、**App log：** Records the details during the calculation process. After completing a calculation, please confirm whether any error messages are output. The presence of any error messages may lead to incorrect calculation results.
Note: In scale mode, the calculated scale factor is also recorded here (indicated by the red box in the figure). Please fill the result into the Scale factor field accordingly.



2.2 Introduction to the wave slice interface functions.

1、**Experiment：** Retrieve the experimental information from the table on the Table input page. Select the experiments that require wavelength check.

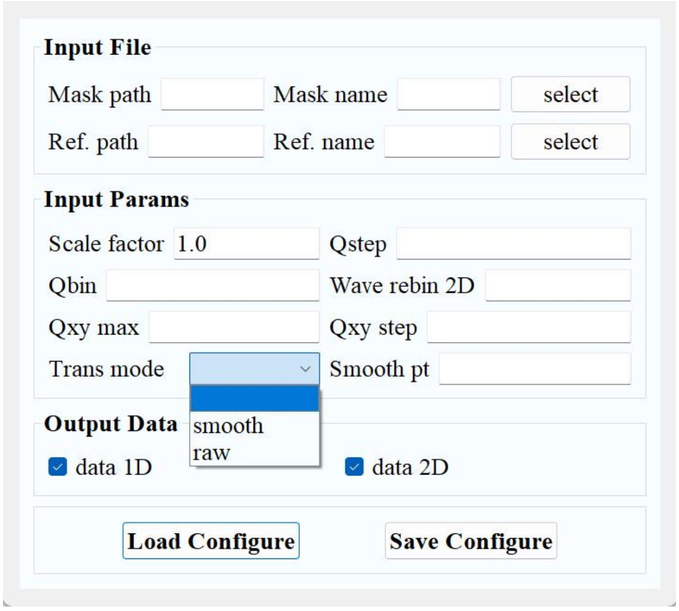


2、Wave slice parameters.

- **add：** Add a segment within the wavelength range. Segments are equally spaced based on the start wavelength of the first segment and the end wavelength of the last segment.
- **delete：** Remove a segment from the wavelength range. Segments are equally spaced based on the start wavelength of the first segment and the end wavelength of the last segment.
- **RUN：** Start the calculation, and save the wave slice results to the "check" folder under the Save path.
- **plot：** Plot segmented images according to the wavelength slicing configuration.

2.3 Reduction parameter control panel.

Access via the Reduction menu: This opens the expert parameter panel, where detailed parameters used in the reduction process can be adjusted.



1、**Mask path:** Click "Select" to choose the mask file you wish to use.

2、**Ref path:** Click "Select" to choose the Ref file you wish to use, which is used for the calculation of the Scale factor.

3、**Trans mode:**

- smooth: Apply smoothing correction to the transmittance curve.
- raw: Use the uncorrected transmittance curve.

4、smooth pt: Number of points for the smoothed transmittance.

5、Save configure: Save the modified results.

6、Qstep: Step size of the 1D scattering curve when converting to q.

7、Qbin: Number of points that make up the 1D scattering curve when converting to q.

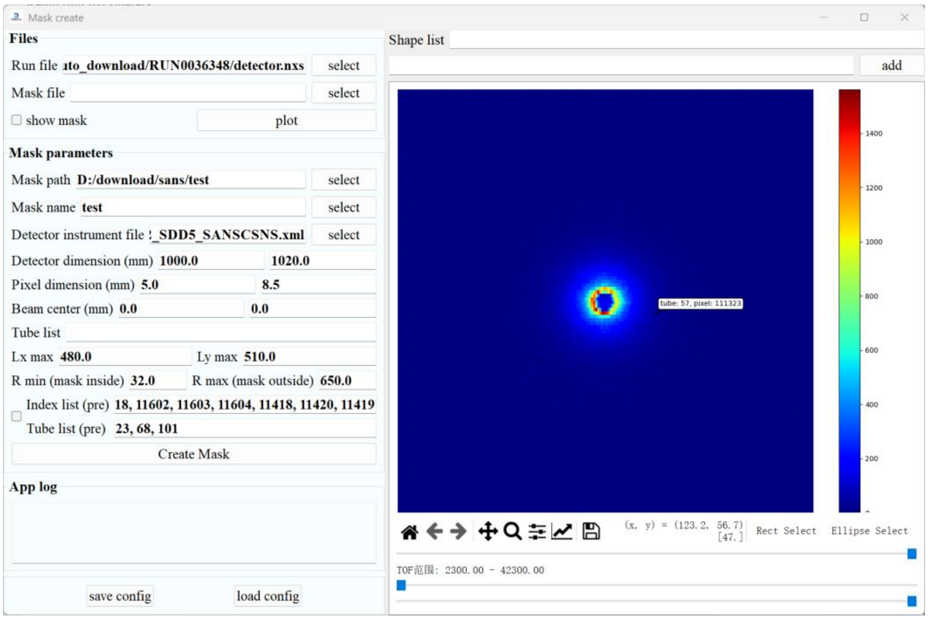
8、wave rebin 2D: Wavelength range for the 2D detector image.

9、Qxy max: Maximum q value of the 2D small-angle scattering image when converting to q.

10、Qxy step: Step size of the 2D small-angle scattering image when converting to q.

2.4 Create a mask file.

Access via the Mask menu: This opens the mask panel, where a mask file can be created based on a specific RUN sample.



1、Files area.:

- Run file: Select the RUN file to be plotted.
- Mask file: Select an existing Mask file (if this is your first time and no mask file is available, you can leave it blank).
- show mask: Whether to plot the mask status.
- Plot: Plot the 2D image.

2、Mask parameters area:

- Mask path: Select the path where the mask file will be saved.
- Mask name: Enter the desired name for the mask file to be saved.
- Detector ins file: Detector parameter file.
- Detector dimension: Actual size of the detector.
- Pixel dimension: Pixel size.
- Beam center: Beam center.
- Tube list: Tube numbers to be masked.
- Lx max and Ly max: The red area is masked. The schematic diagram is shown below.
- R min and R max: With the beam center as the origin, mask the regions inside R min and outside R max.
- Index list (pre): Predefined pixels to be masked.
- Tube list (pre): Predefined tubes to be masked (check this if modifications are needed).
- Create Mask: Start creating and saving the mask file.

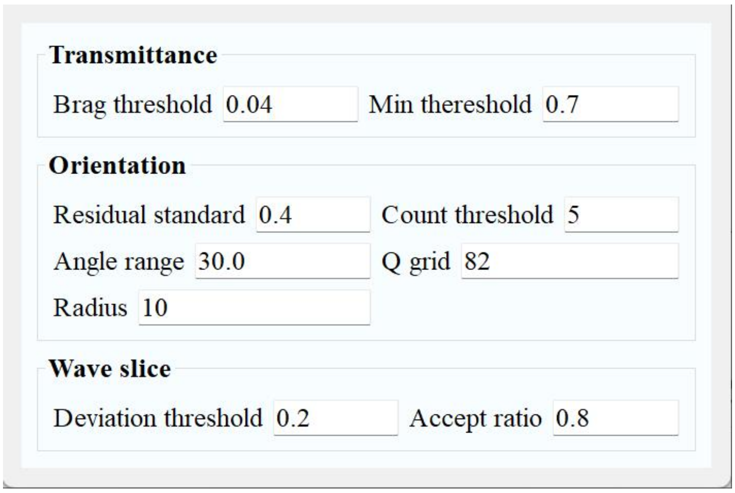
3、Plot area:

- Rect Select: Drag the mouse to select a rectangular area to be masked.
- Ellipse Select: Drag the mouse to select an elliptical area to be masked.
- In the upper-right area, the regions currently selected using Rect Select and Ellipse Select are updated in real time. Click "Add" to add the selected region to the Shape list.
- Shape list: A summary of custom rectangular and elliptical regions to be masked.
- Adjust Slider 1: Change the maximum counts value to enhance the visual appearance of bad pixels, assisting in mask creation.

- Adjust Sliders 2 and 3: Change the TOF summation range to help identify anomalous scattering signals.

2.5 Automatic reduction check function control panel.

Access via the Check menu: This opens the Check param panel, where the criteria for checking Bragg edges and orientation can be adjusted.



1、Transmittance: Determines the presence of Bragg edges based on peak values and the lower limit of transmittance.

- Bragg threshold: A quadratic curve is fitted to the transmittance data. If the goodness of fit is greater than the Bragg threshold, the data is considered to have no Bragg edge. The larger the Bragg threshold is set, the less likely a Bragg edge will be overlooked.
- Min threshold: Sets the minimum allowable value for transmittance.

2、Orientation: Constructs an intensity-versus-angle curve along an arc centered on the beam spot, and determines the presence of orientation based on peak detection.

- Residual standard: A first-order fit is performed on the intensity-versus-angle curve along an arc centered on the beam spot, and the residuals are calculated. The residuals are normalized by the fitted curve. If there are peaks exceeding the Residual standard, the data is considered to have orientation. The larger this value is set, the less likely the data will be judged as having orientation.
- Count threshold: Excludes the influence of a small number of outlier points on the judgment result. Outliers exceeding the Residual standard must be greater than the Count threshold to be judged as indicating the presence of orientation.
- Angle range: The integration width along the angle direction determined empirically. This represents the angular sector over which integration is performed along the orientation direction to calculate the 1D Iq curve.
- Q grid: The number of interpolation points used when interpolating the 2D scattering pattern.
- Radius: The radius of the arc used for judgment, centered on the beam spot. The unit is dimensionless and represents the number of pixels.

3、Wave slice

- Deviation threshold: Criterion for outliers, representing the percentage deviation from the average of the current reference curve.

- Accept ratio: If the number of inlier points as a percentage of the total points exceeds the Accept ratio, the result is considered to have good agreement.

2.6 Introduction to the reduction results.